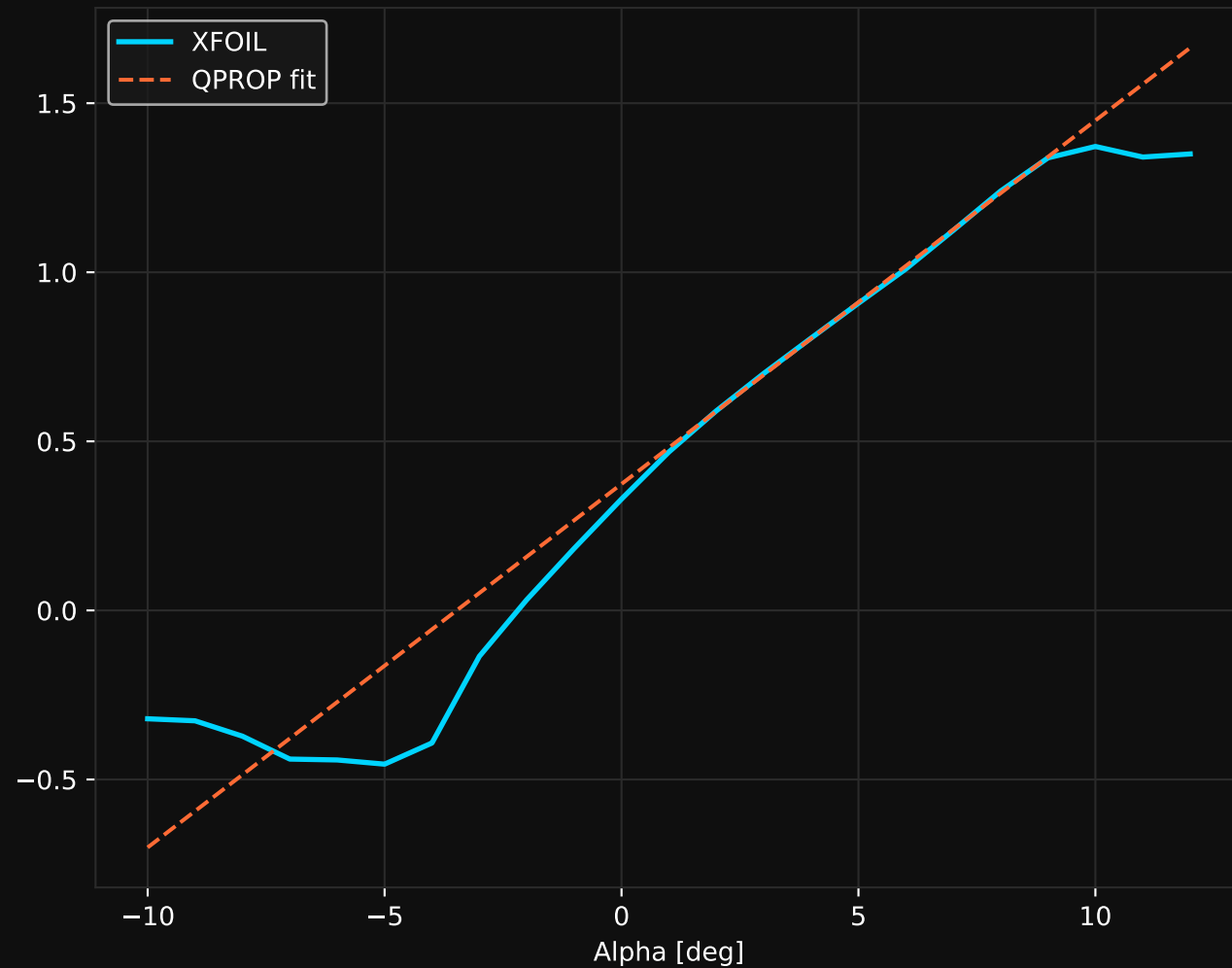
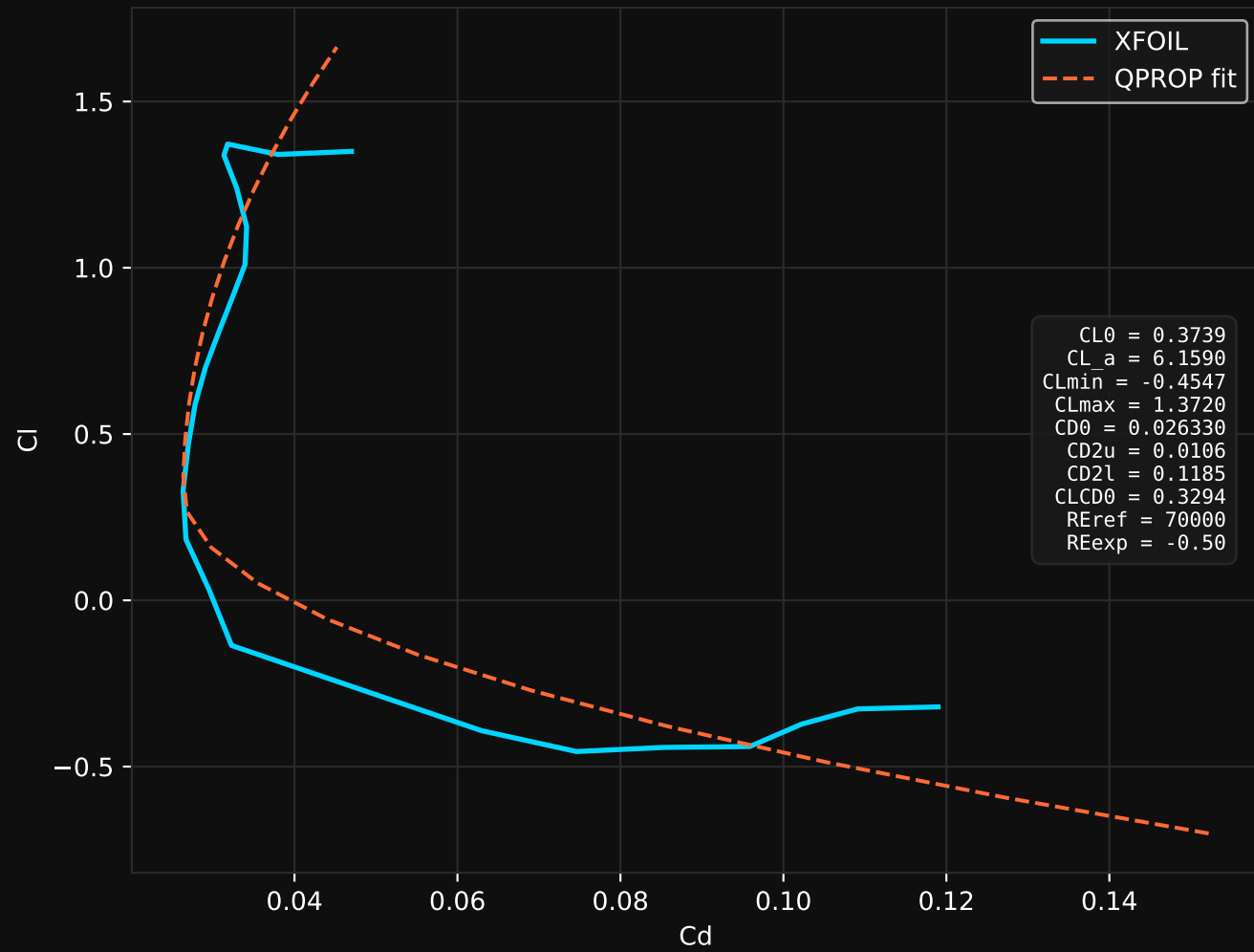


# Airfoil: NACA4412 | Re = 70000

## Cl x Alpha

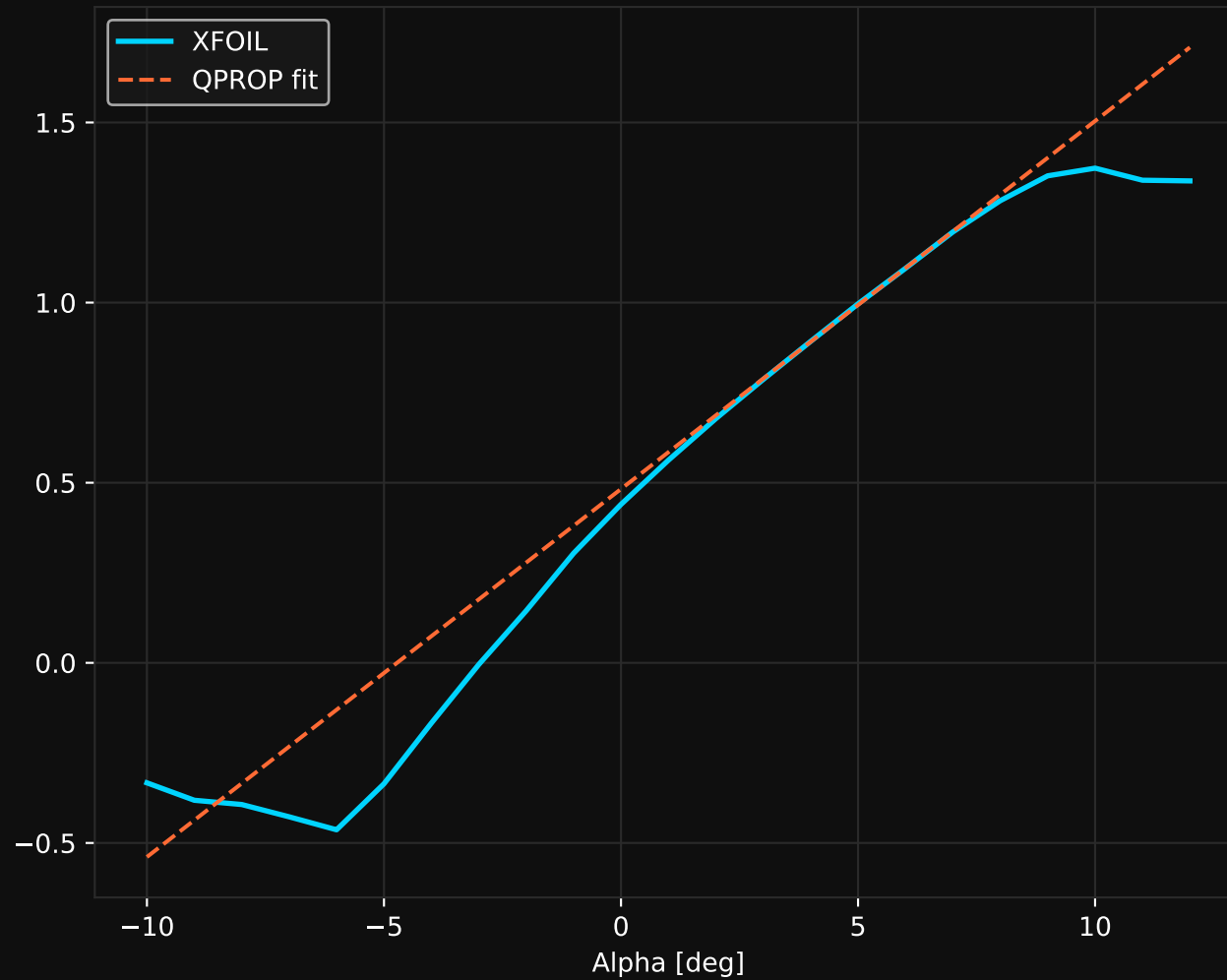


## Cl x Cd (polar)

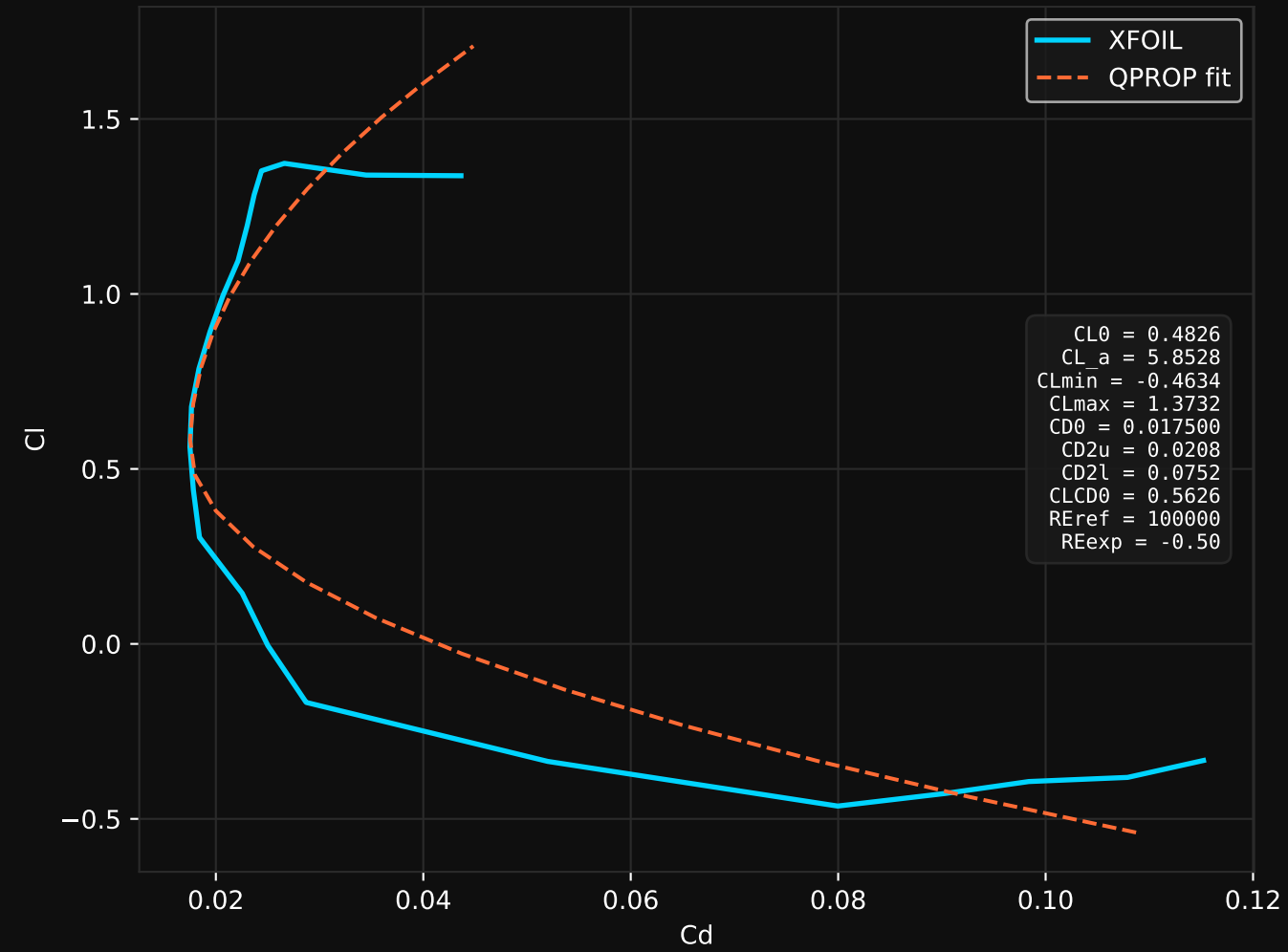


# Airfoil: NACA4412 | $Re = 100000$

$C_l$  x Alpha



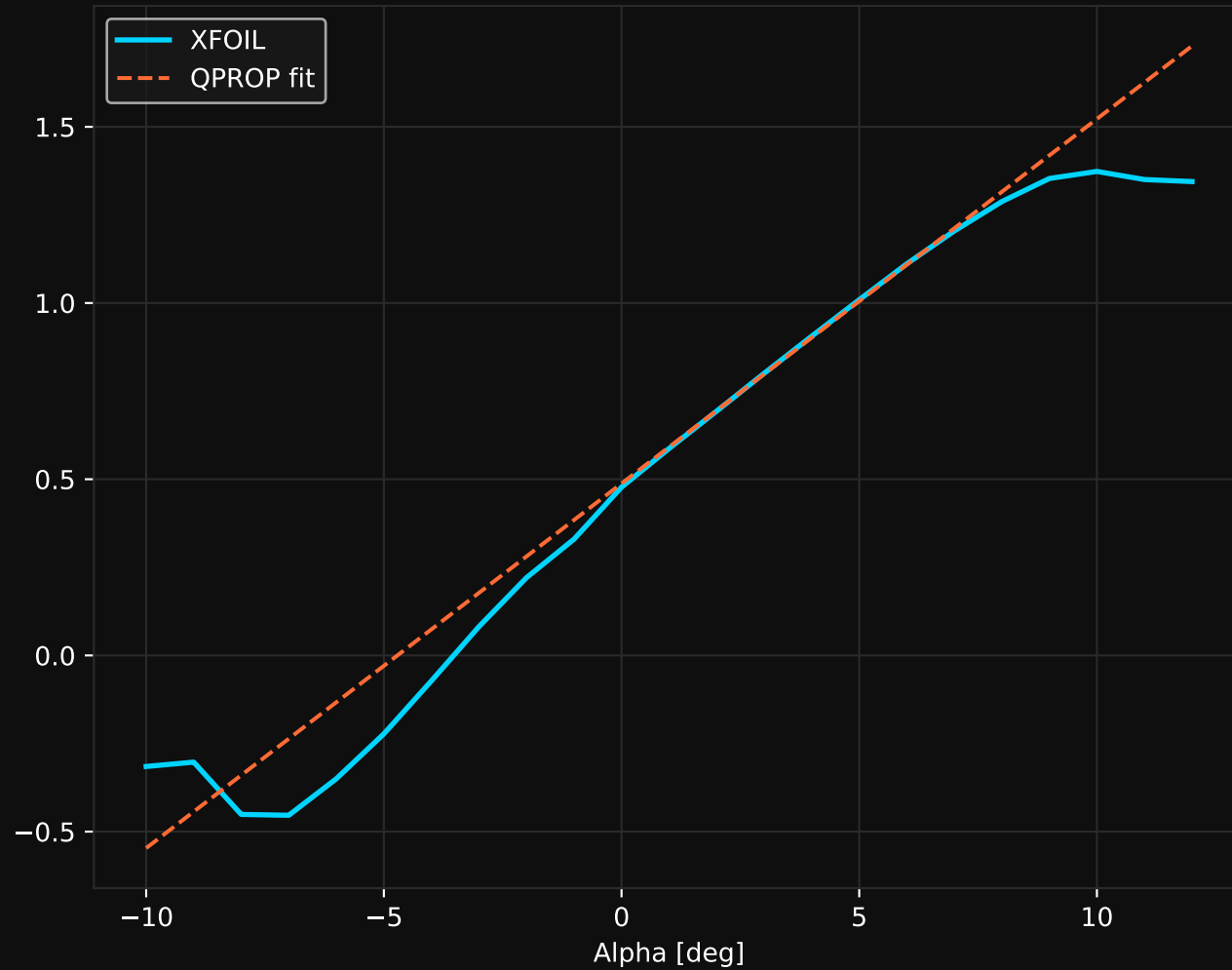
$C_l$  x  $C_d$  (polar)



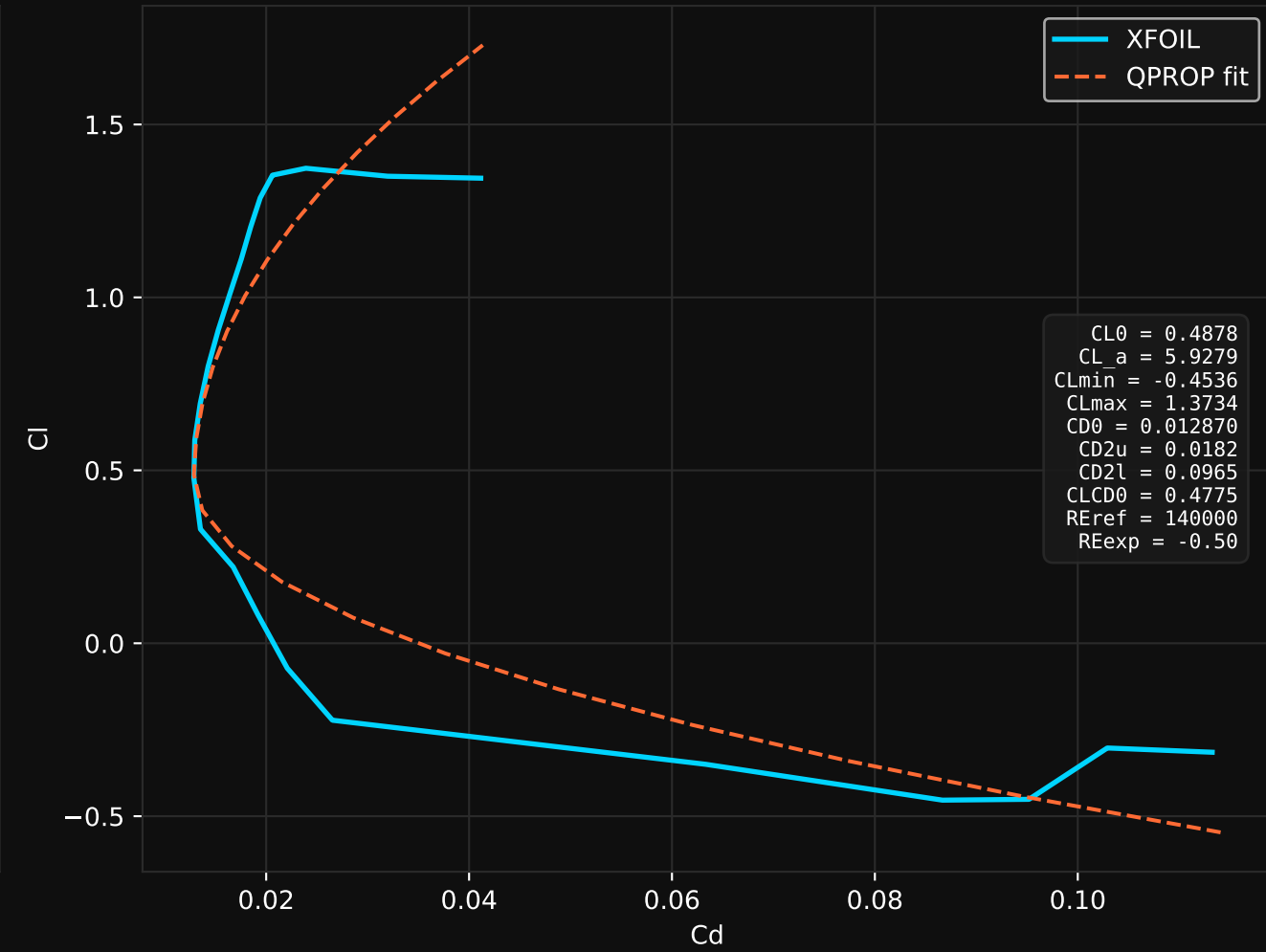
$C_{l0} = 0.4826$   
 $C_{l\alpha} = 5.8528$   
 $C_{lmin} = -0.4634$   
 $C_{lmax} = 1.3732$   
 $C_{D0} = 0.017500$   
 $C_{D2u} = 0.0208$   
 $C_{D2l} = 0.0752$   
 $C_{LCD0} = 0.5626$   
 $Re_{ref} = 100000$   
 $Re_{exp} = -0.50$

# Airfoil: NACA4412 | Re = 140000

Cl x Alpha



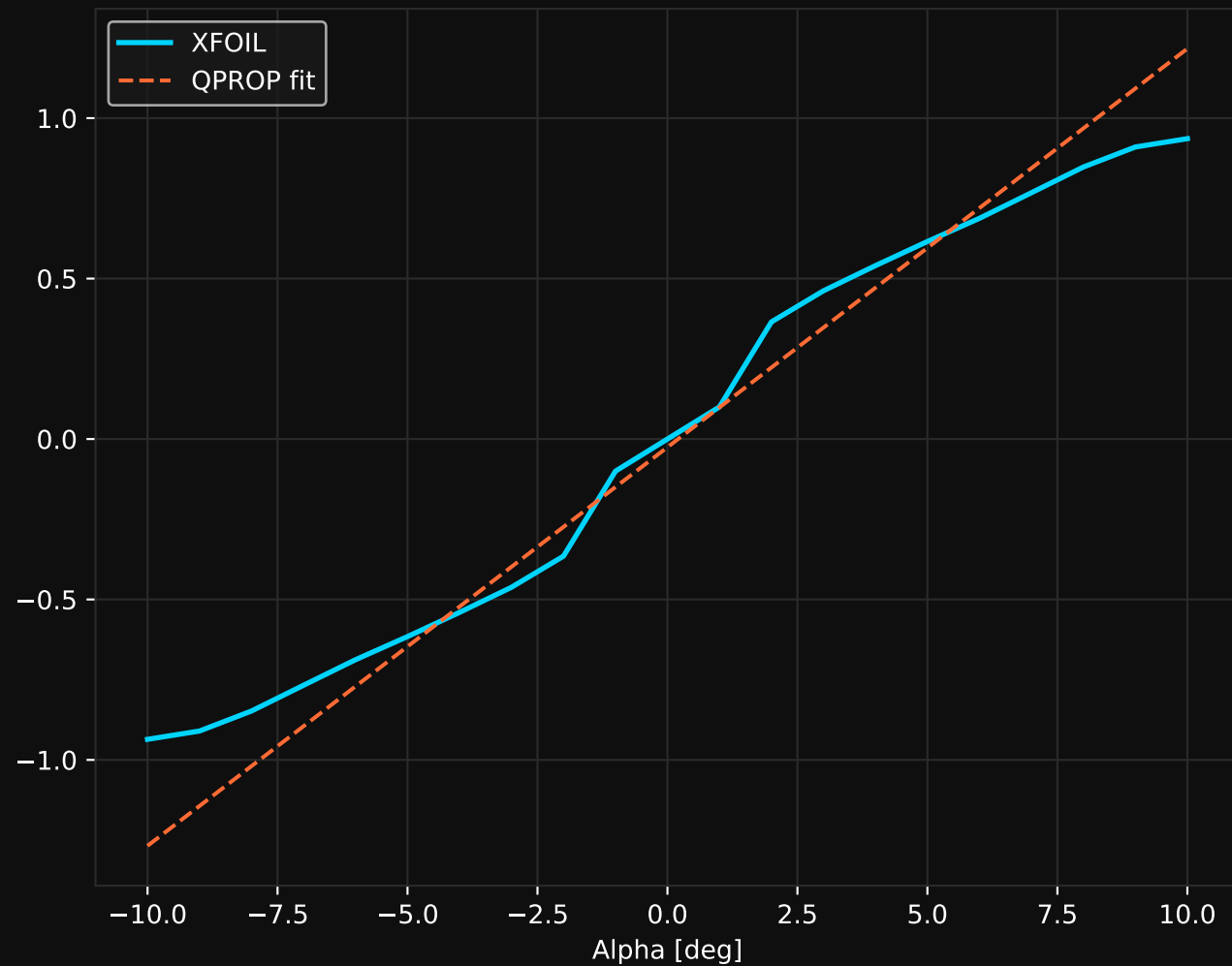
Cl x Cd (polar)



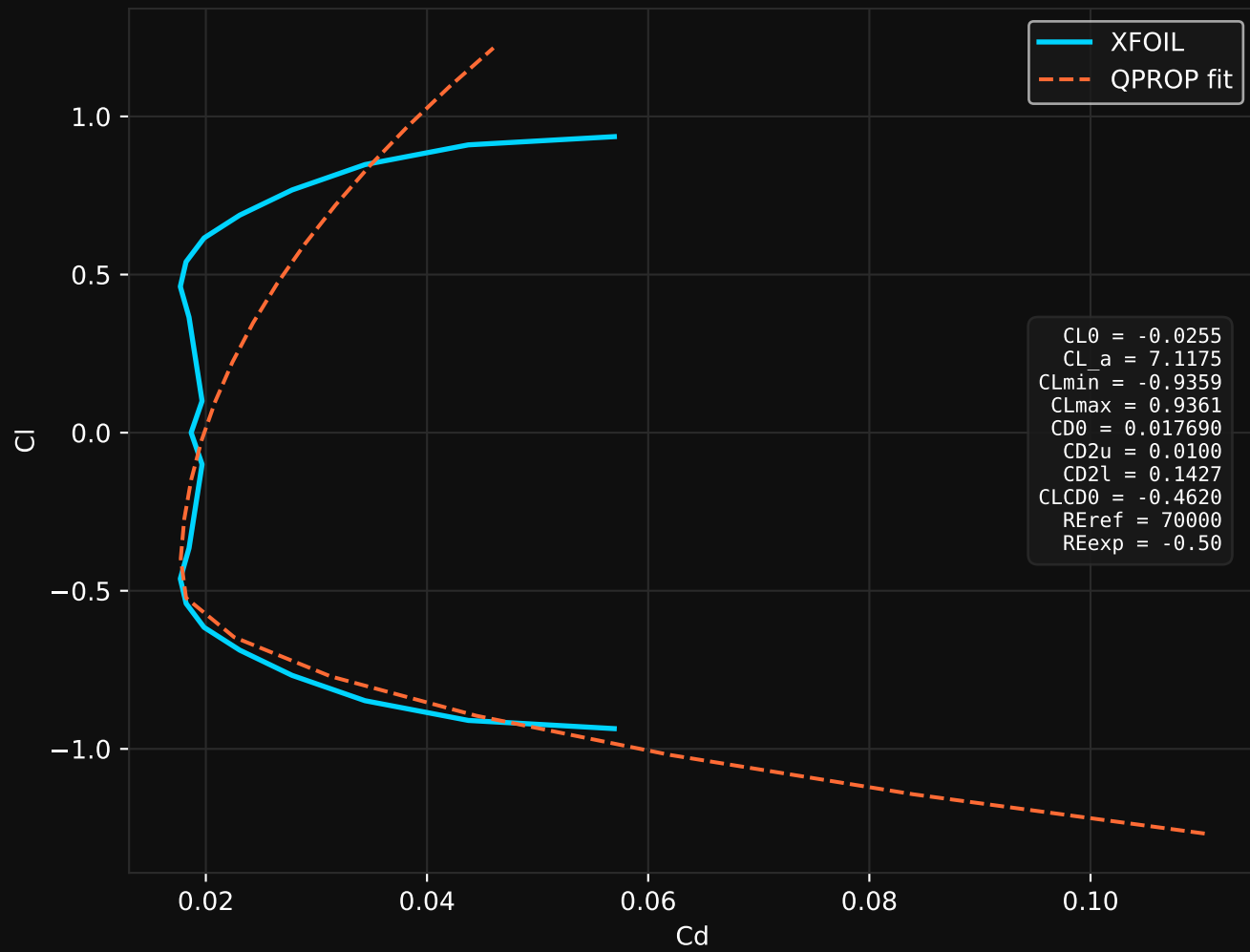
$CL_0 = 0.4878$   
 $CL_a = 5.9279$   
 $CL_{min} = -0.4536$   
 $CL_{max} = 1.3734$   
 $CD_0 = 0.012870$   
 $CD_{2u} = 0.0182$   
 $CD_{2l} = 0.0965$   
 $CLCD_0 = 0.4775$   
 $RE_{ref} = 140000$   
 $RE_{exp} = -0.50$

# Airfoil: NACA0012 | Re = 70000

Cl x Alpha

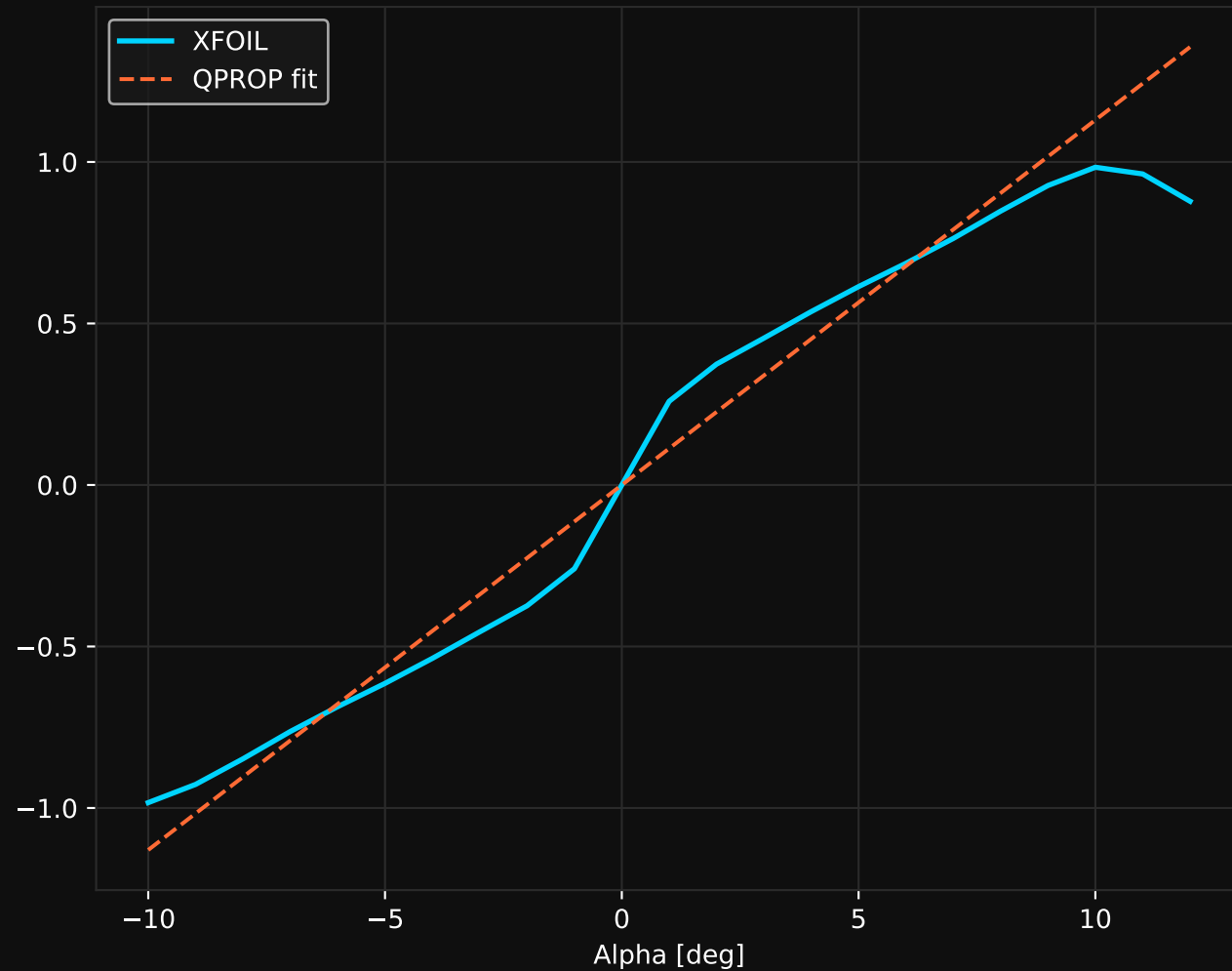


Cl x Cd (polar)

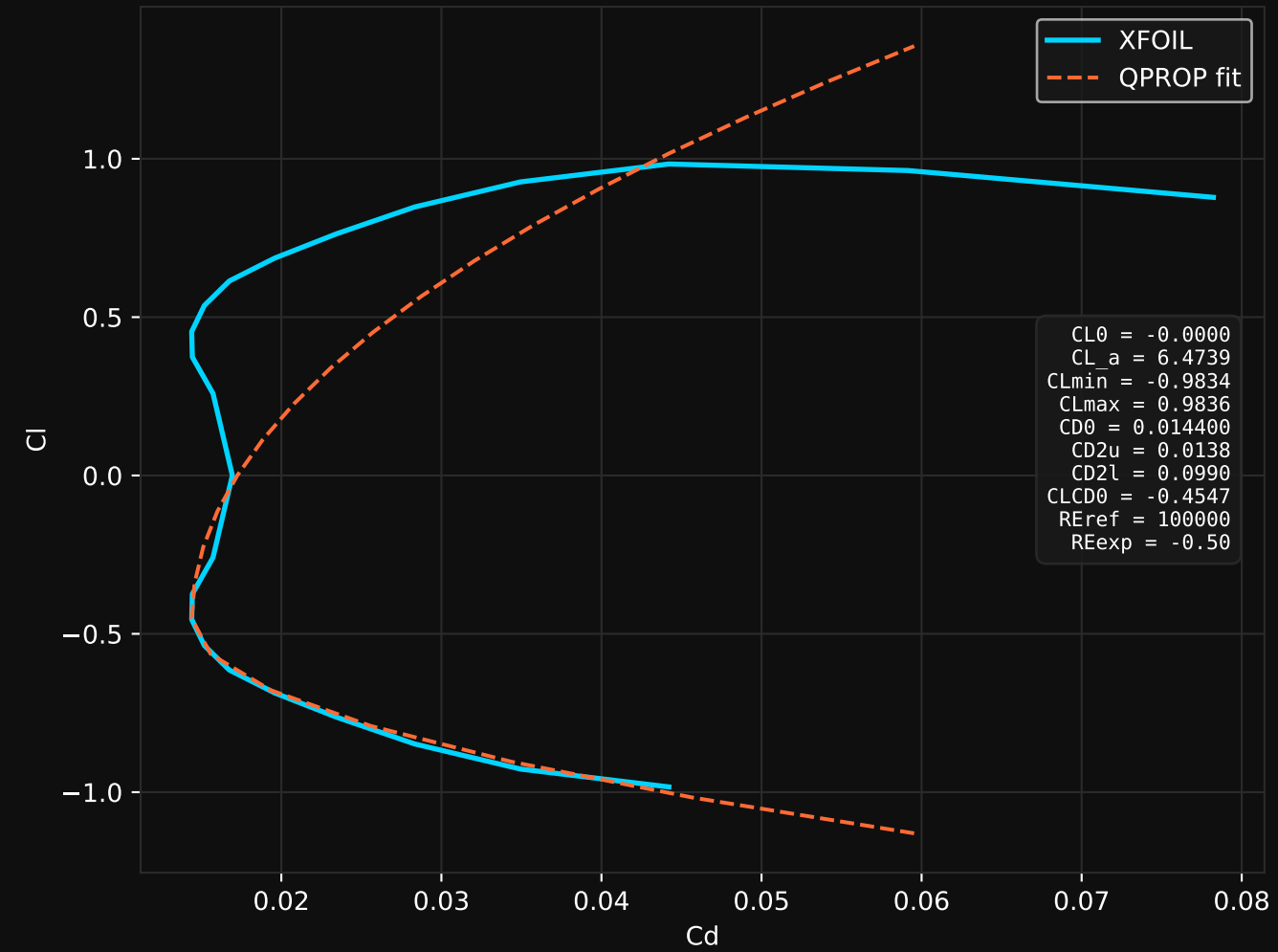


# Airfoil: NACA0012 | $Re = 100000$

$C_l$  x Alpha

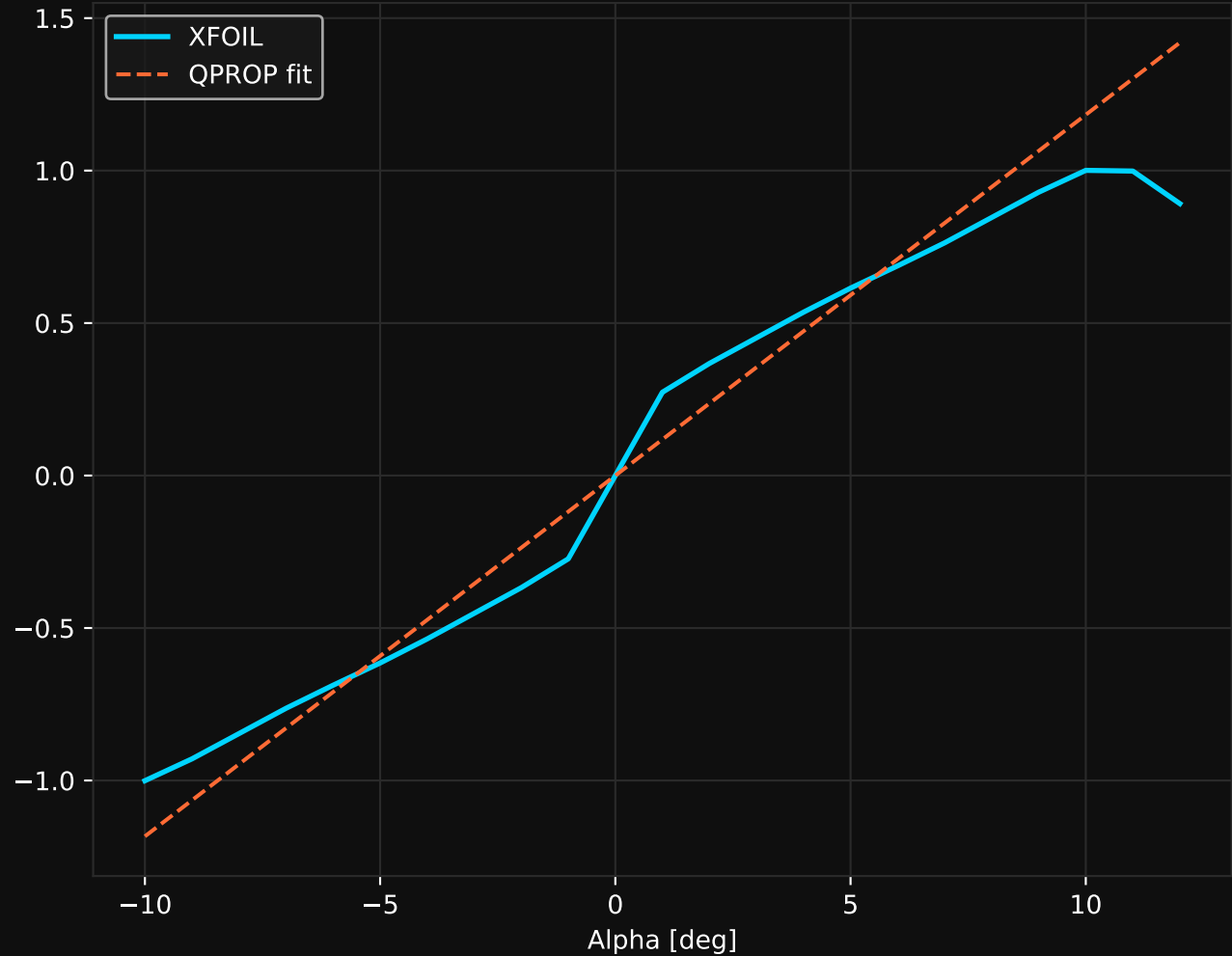


$C_l$  x  $C_d$  (polar)

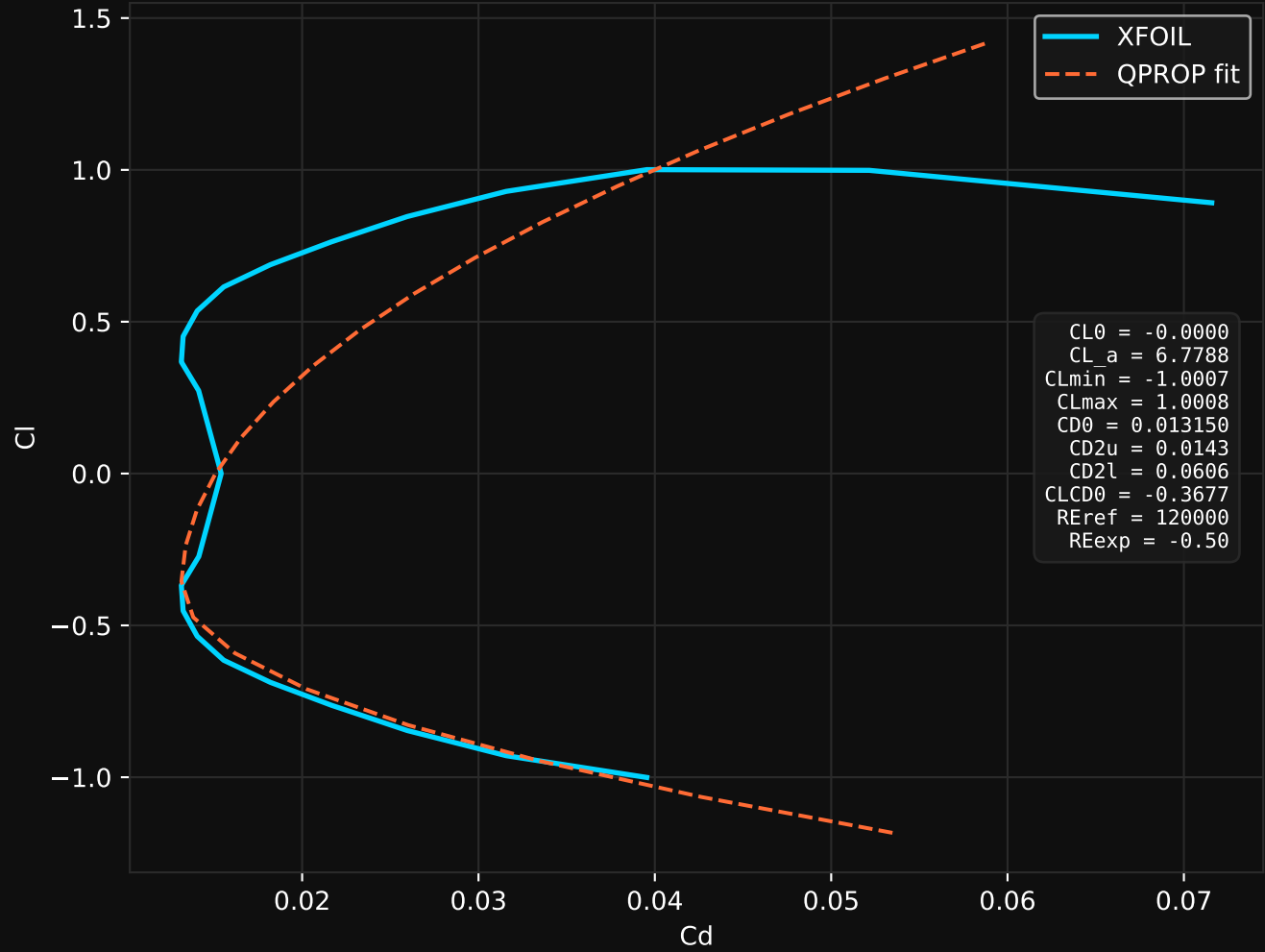


# Airfoil: NACA0012 | Re = 120000

Cl x Alpha

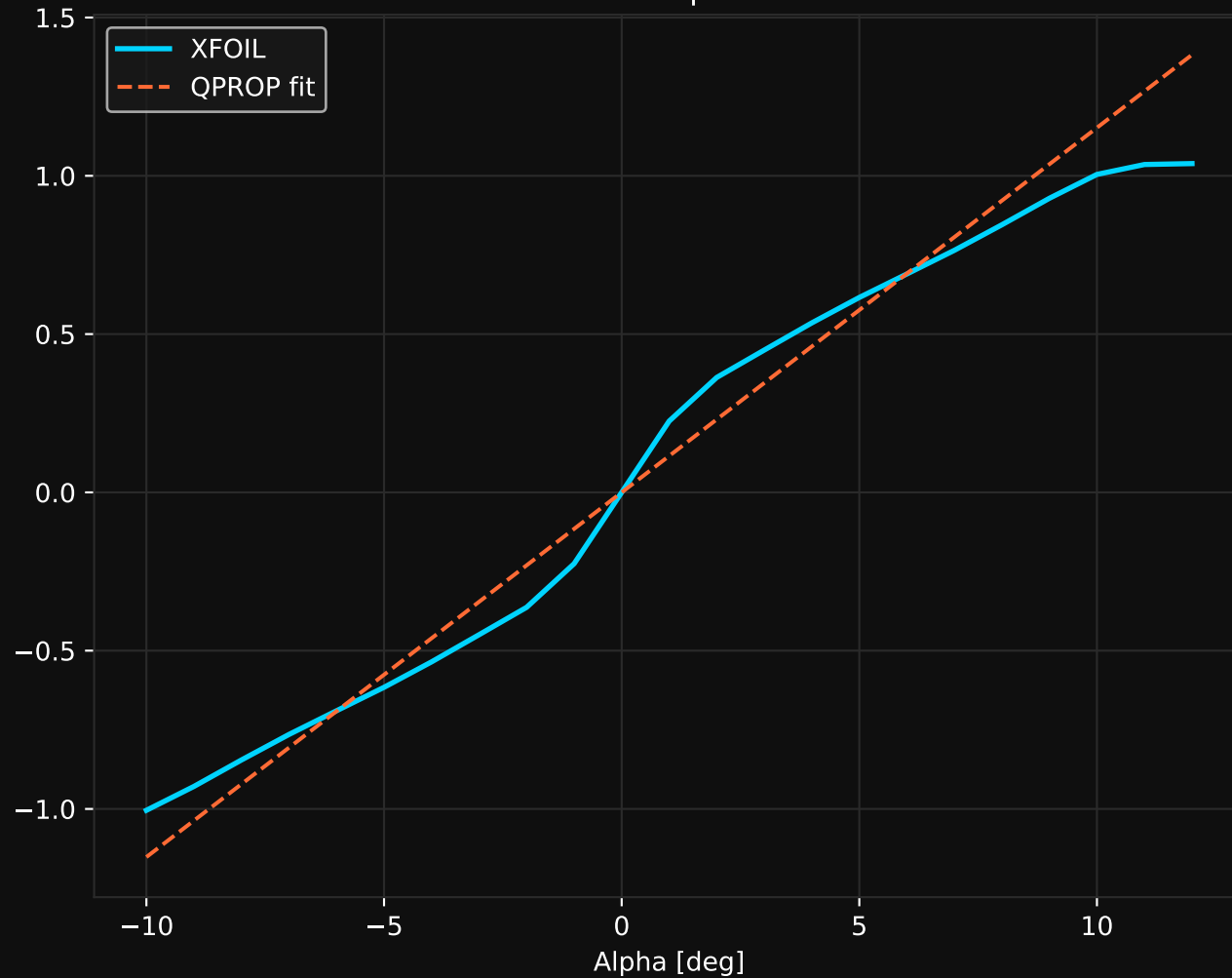


Cl x Cd (polar)

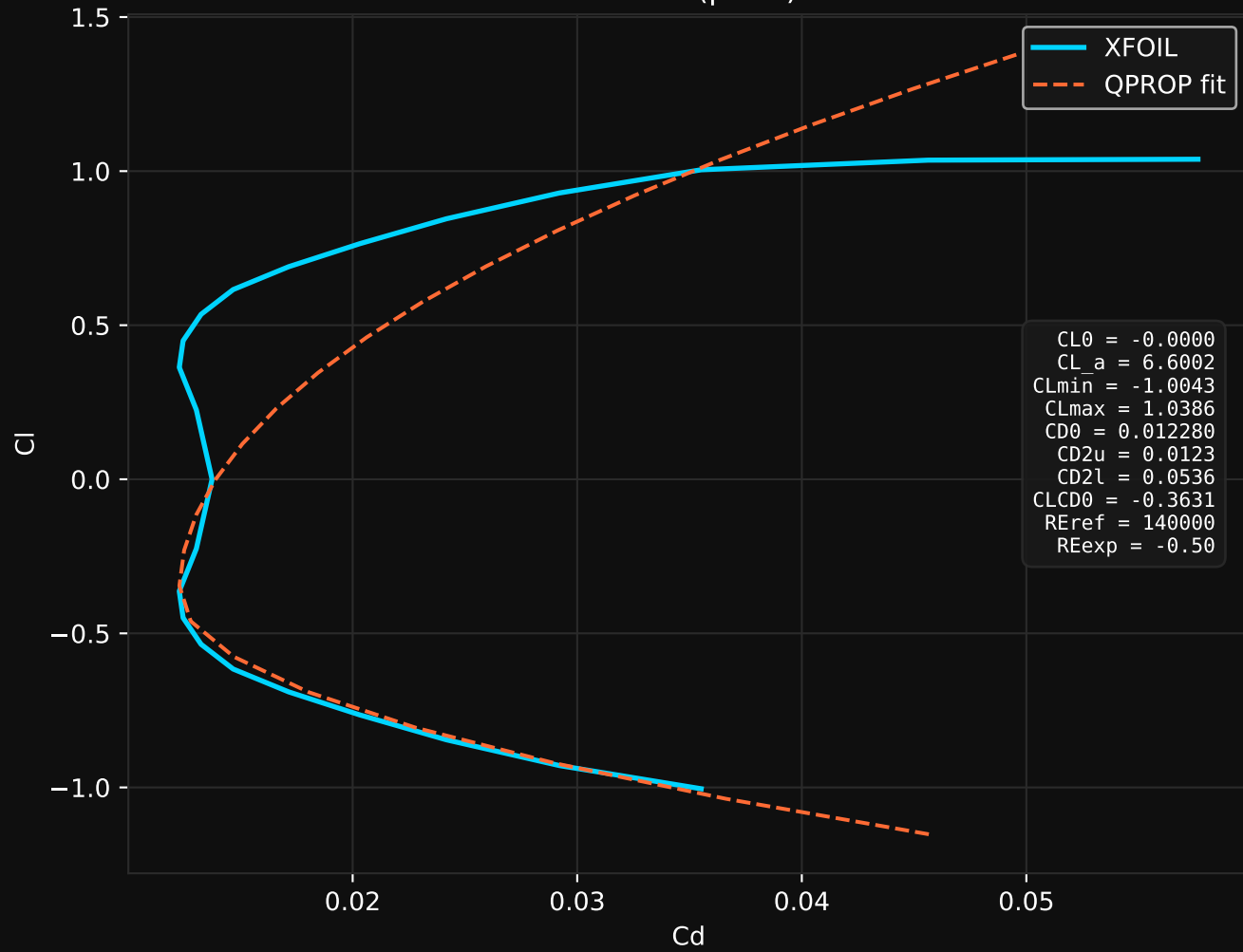


# Airfoil: NACA0012 | Re = 140000

Cl x Alpha

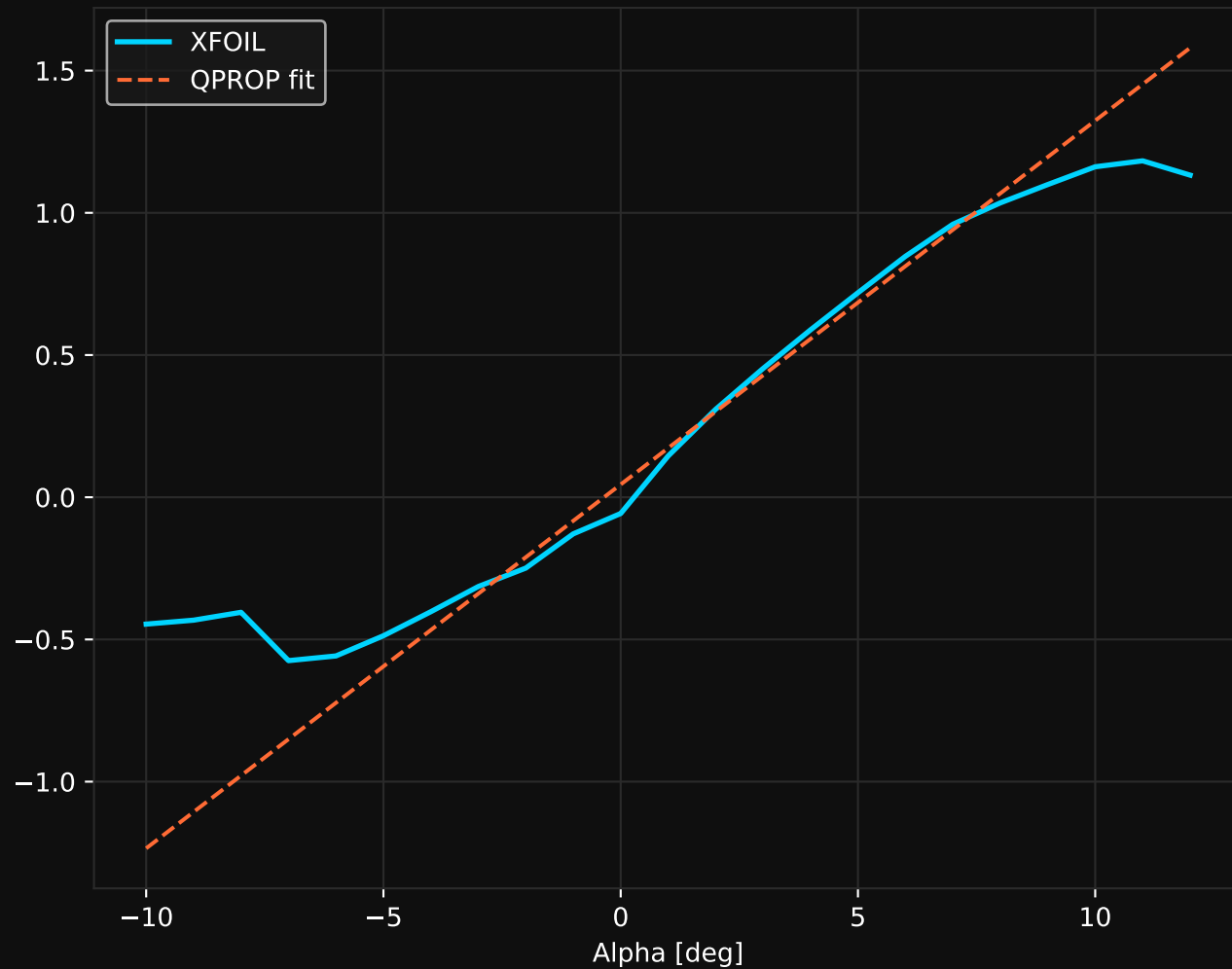


Cl x Cd (polar)

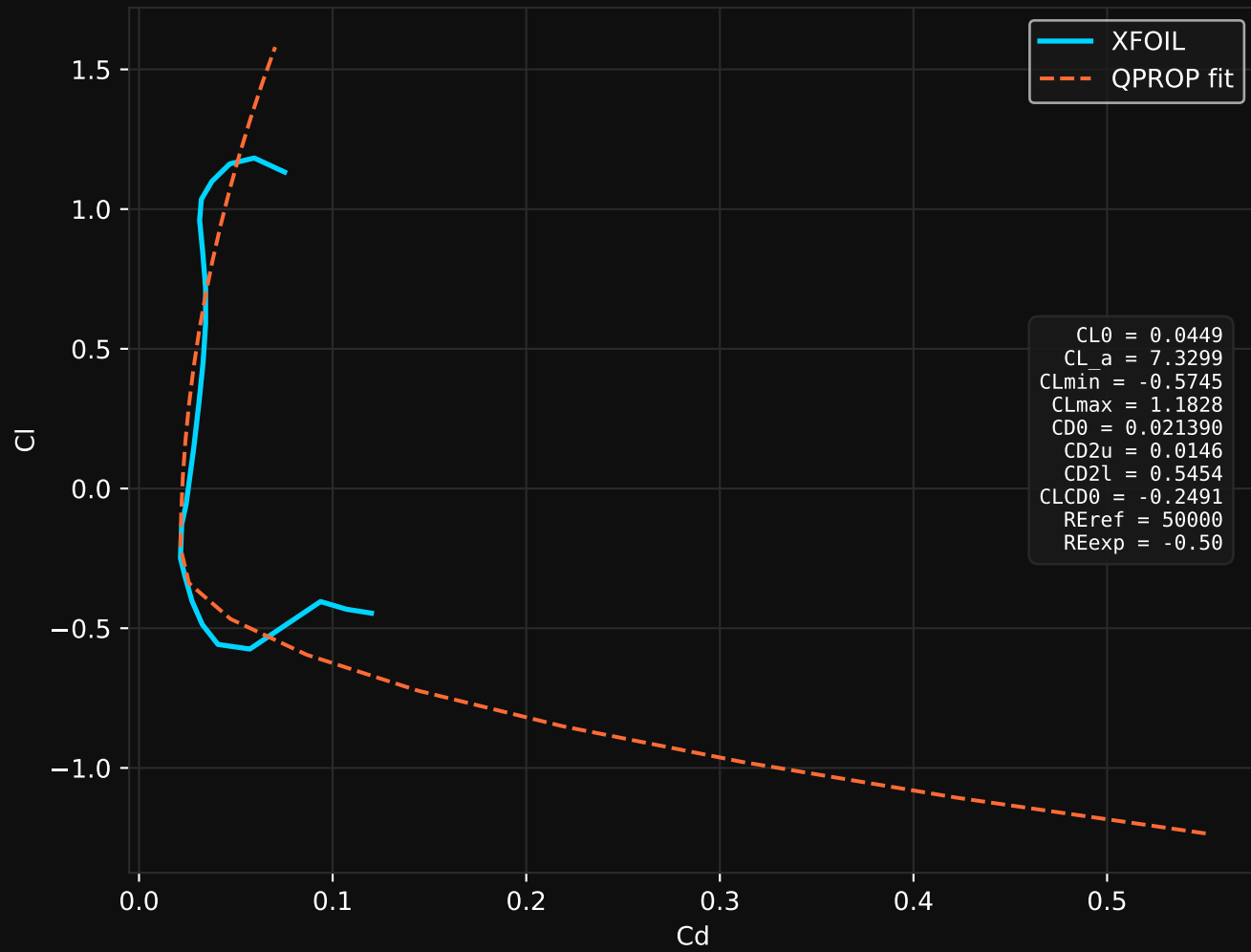


# Airfoil: NACA2412 | Re = 50000

Cl x Alpha



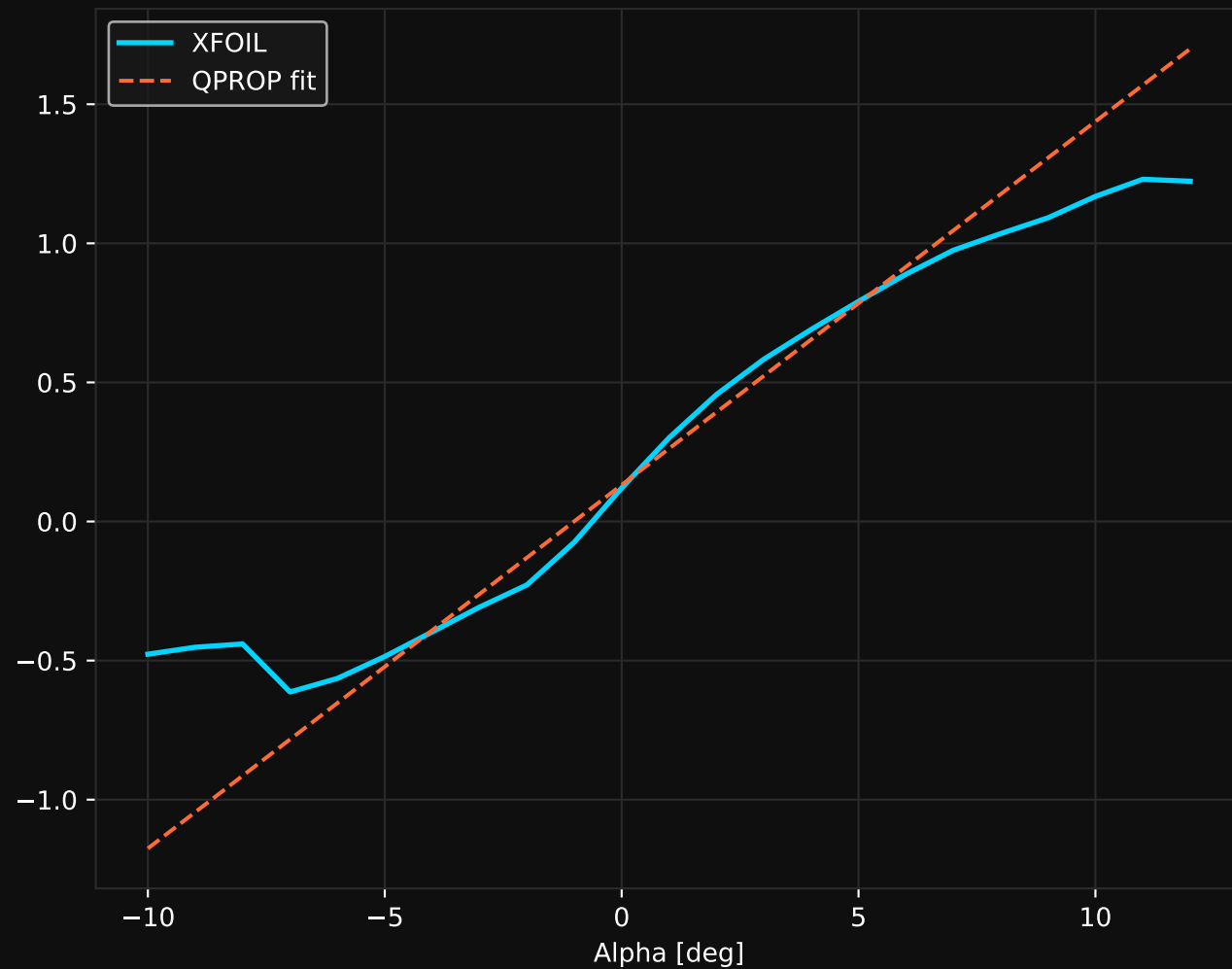
Cl x Cd (polar)



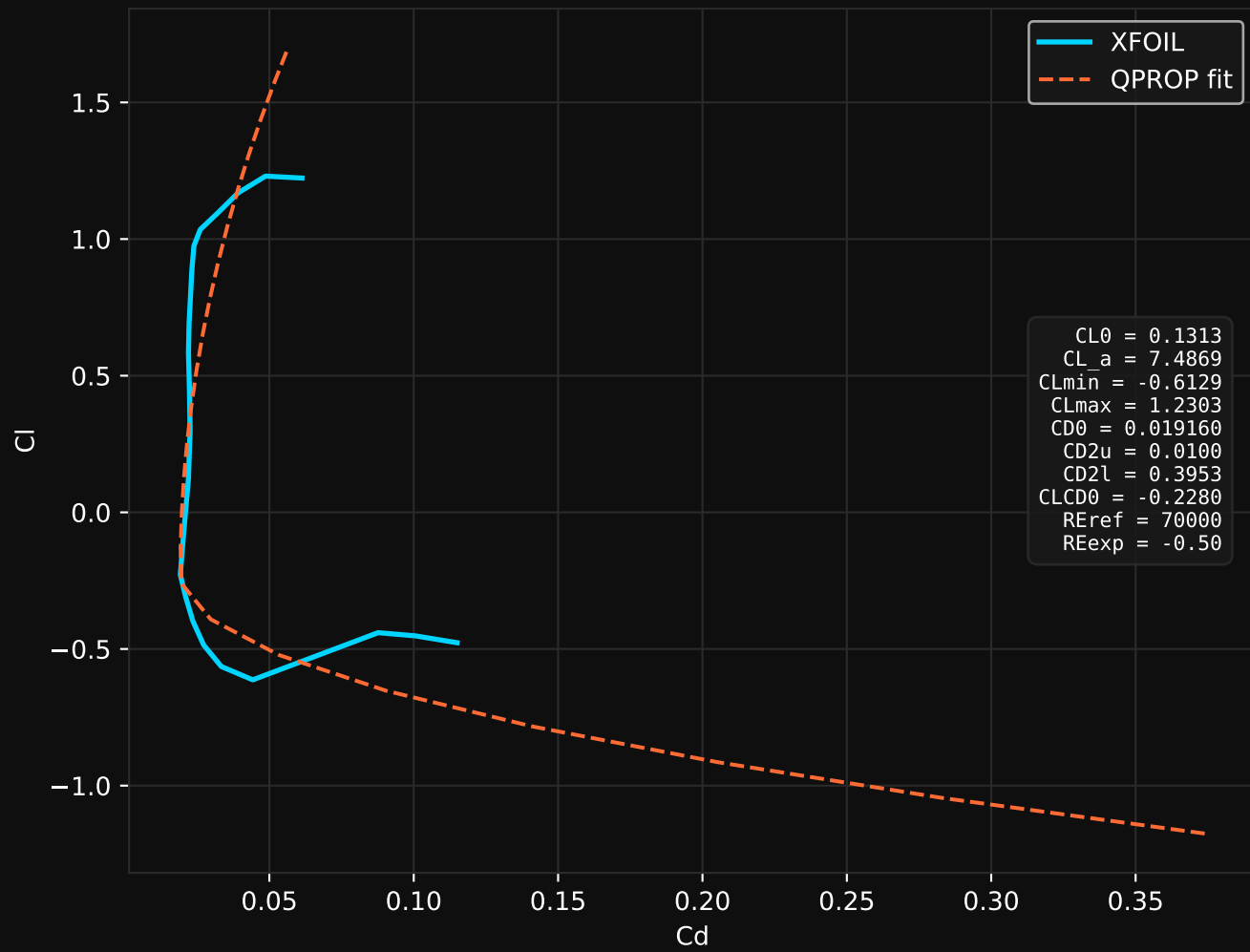


# Airfoil: NACA2412 | Re = 70000

Cl x Alpha

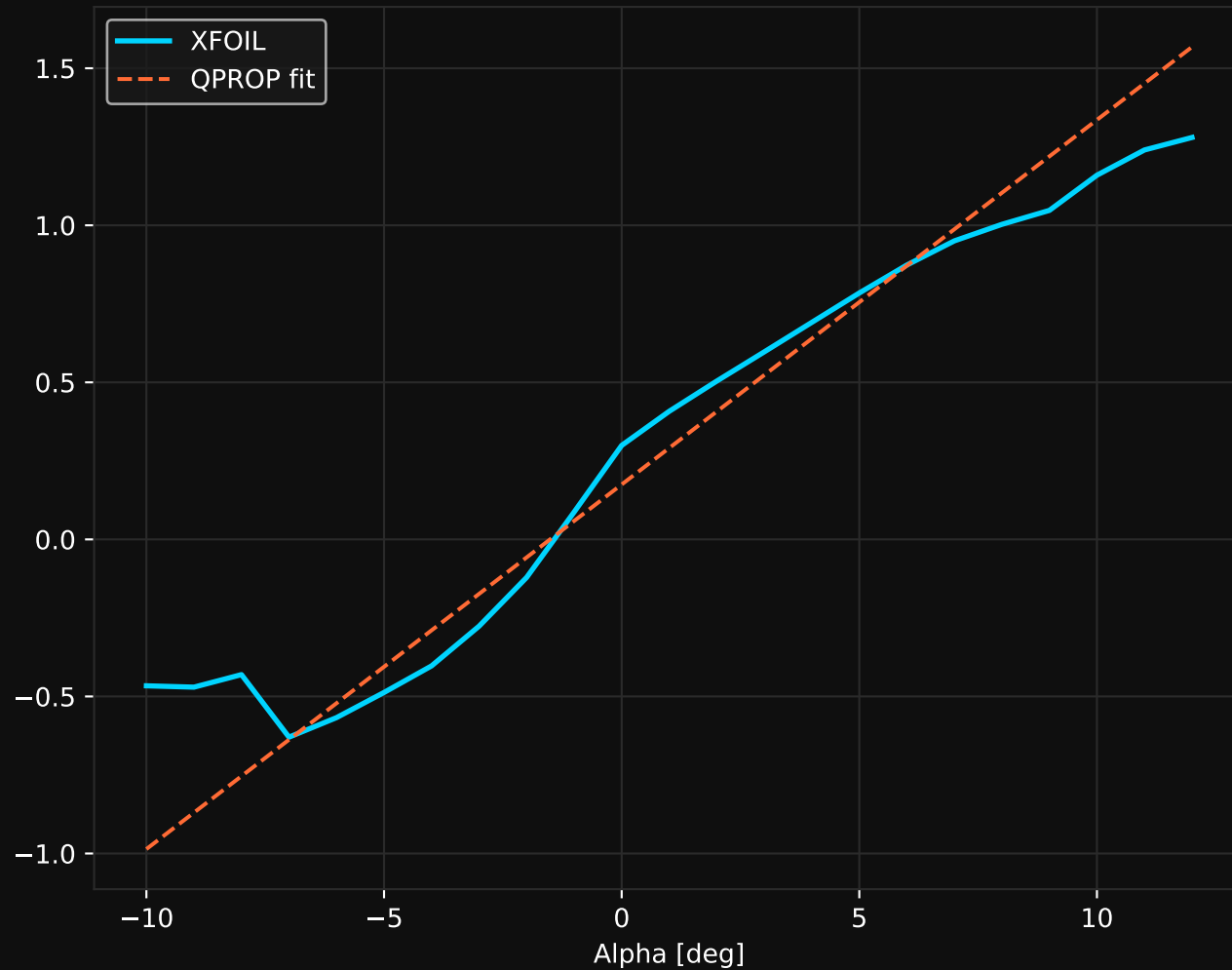


Cl x Cd (polar)

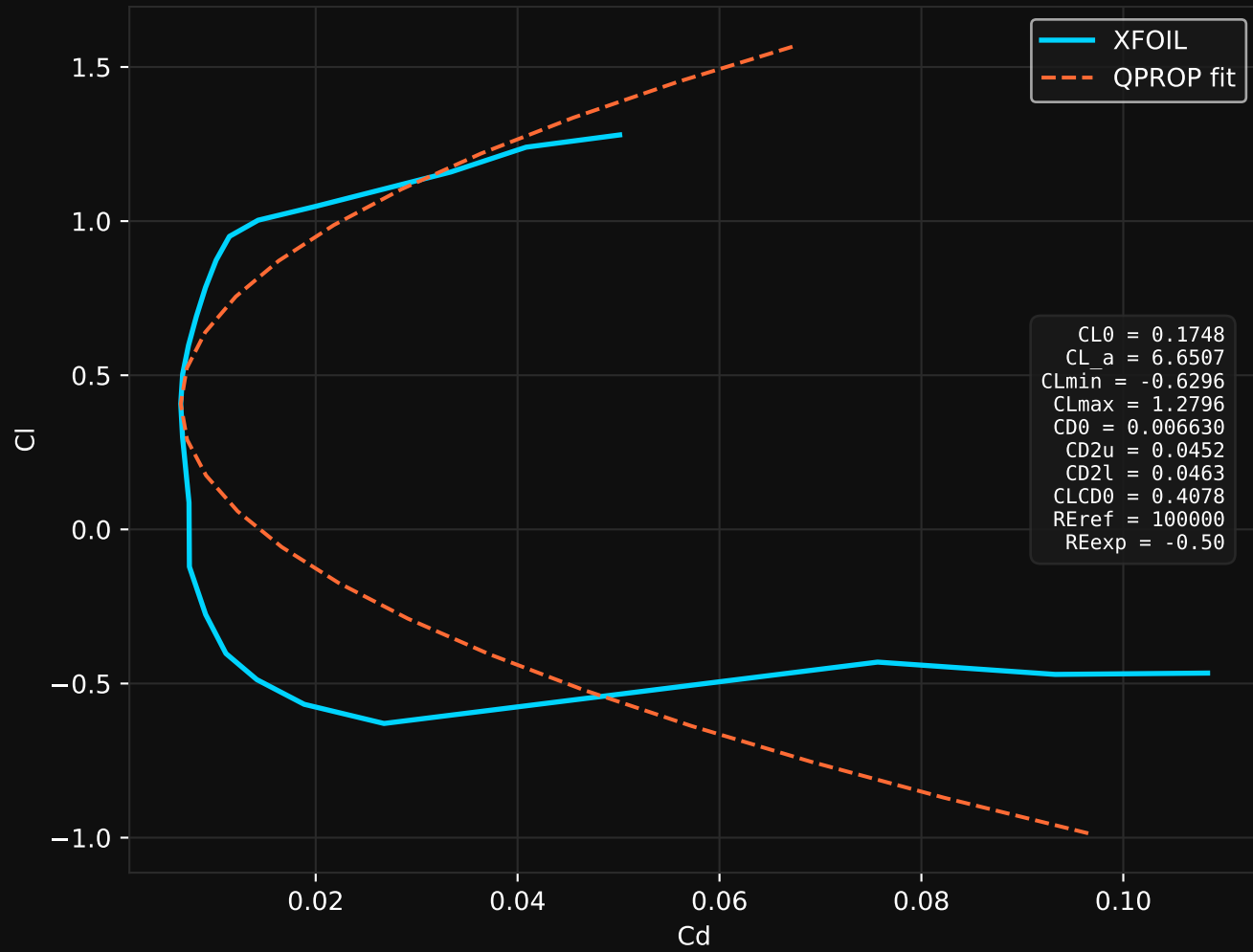


# Airfoil: NACA2412 | Re = 100000

Cl x Alpha

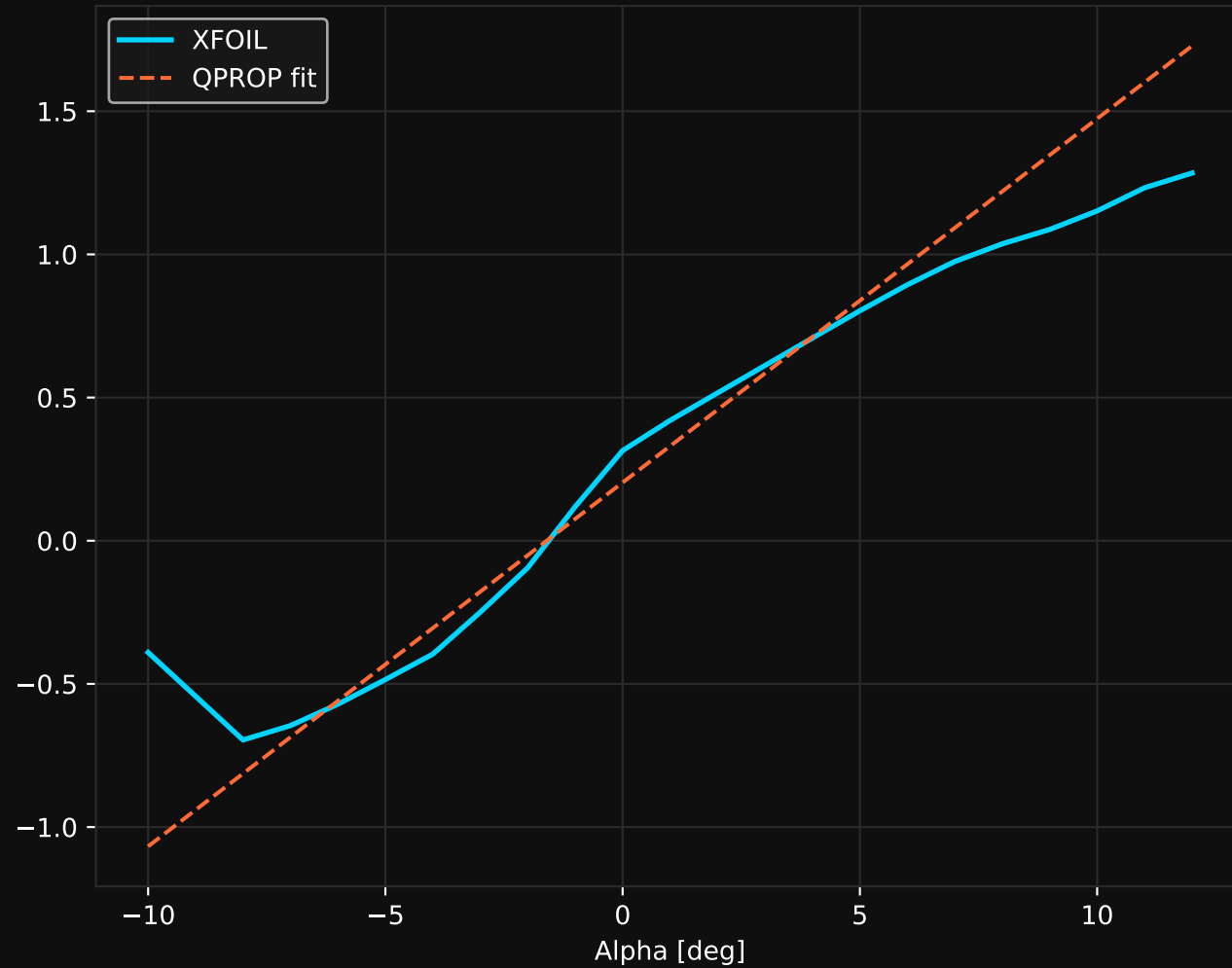


Cl x Cd (polar)

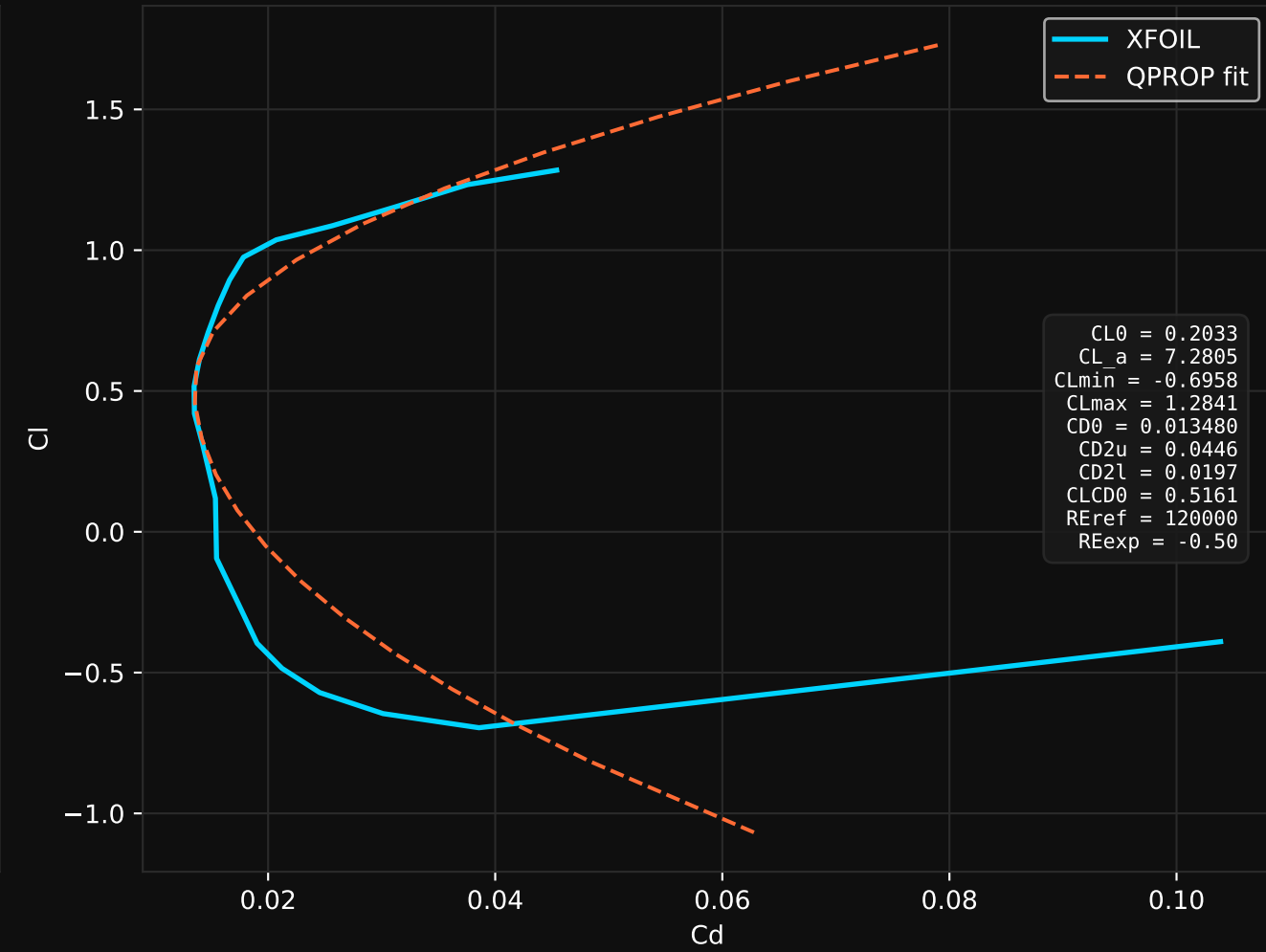


# Airfoil: NACA2412 | Re = 120000

Cl x Alpha



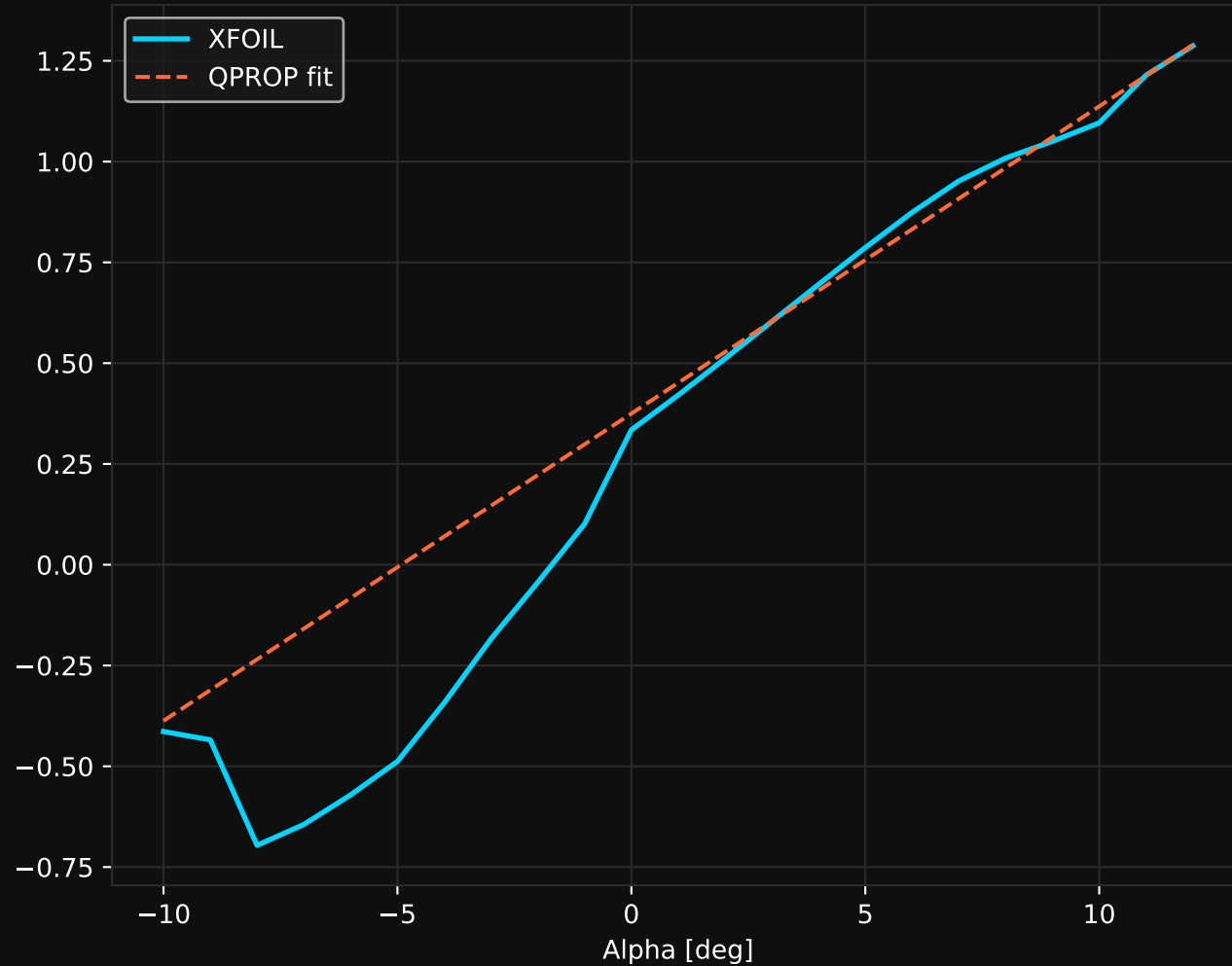
Cl x Cd (polar)



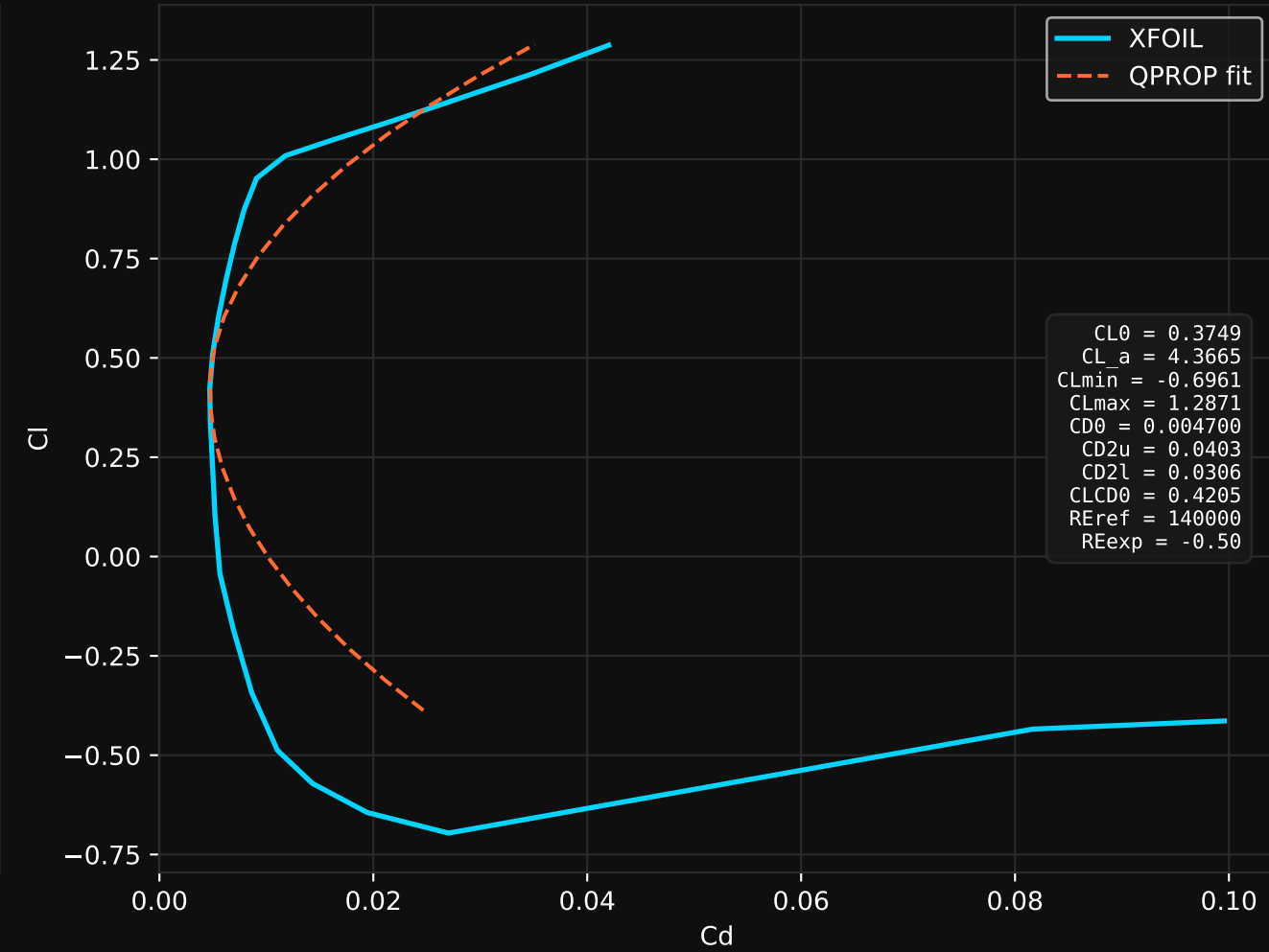
$C_{l0} = 0.2033$   
 $C_{l_a} = 7.2805$   
 $C_{lmin} = -0.6958$   
 $C_{lmax} = 1.2841$   
 $C_{D0} = 0.013480$   
 $C_{D2u} = 0.0446$   
 $C_{D2l} = 0.0197$   
 $C_{LCD0} = 0.5161$   
 $Re_{ref} = 120000$   
 $Re_{exp} = -0.50$

# Airfoil: NACA2412 | Re = 140000

## Cl x Alpha

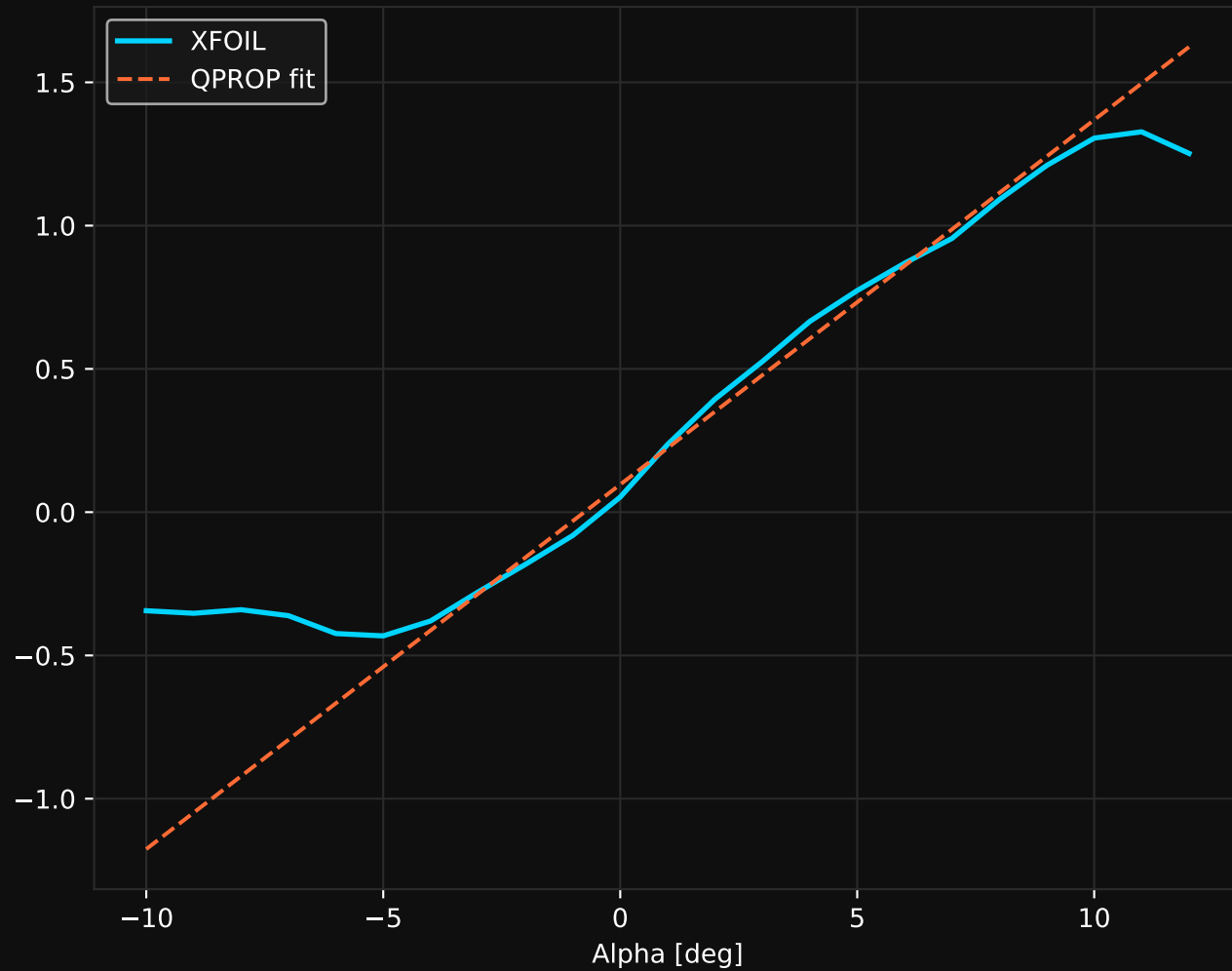


## Cl x Cd (polar)

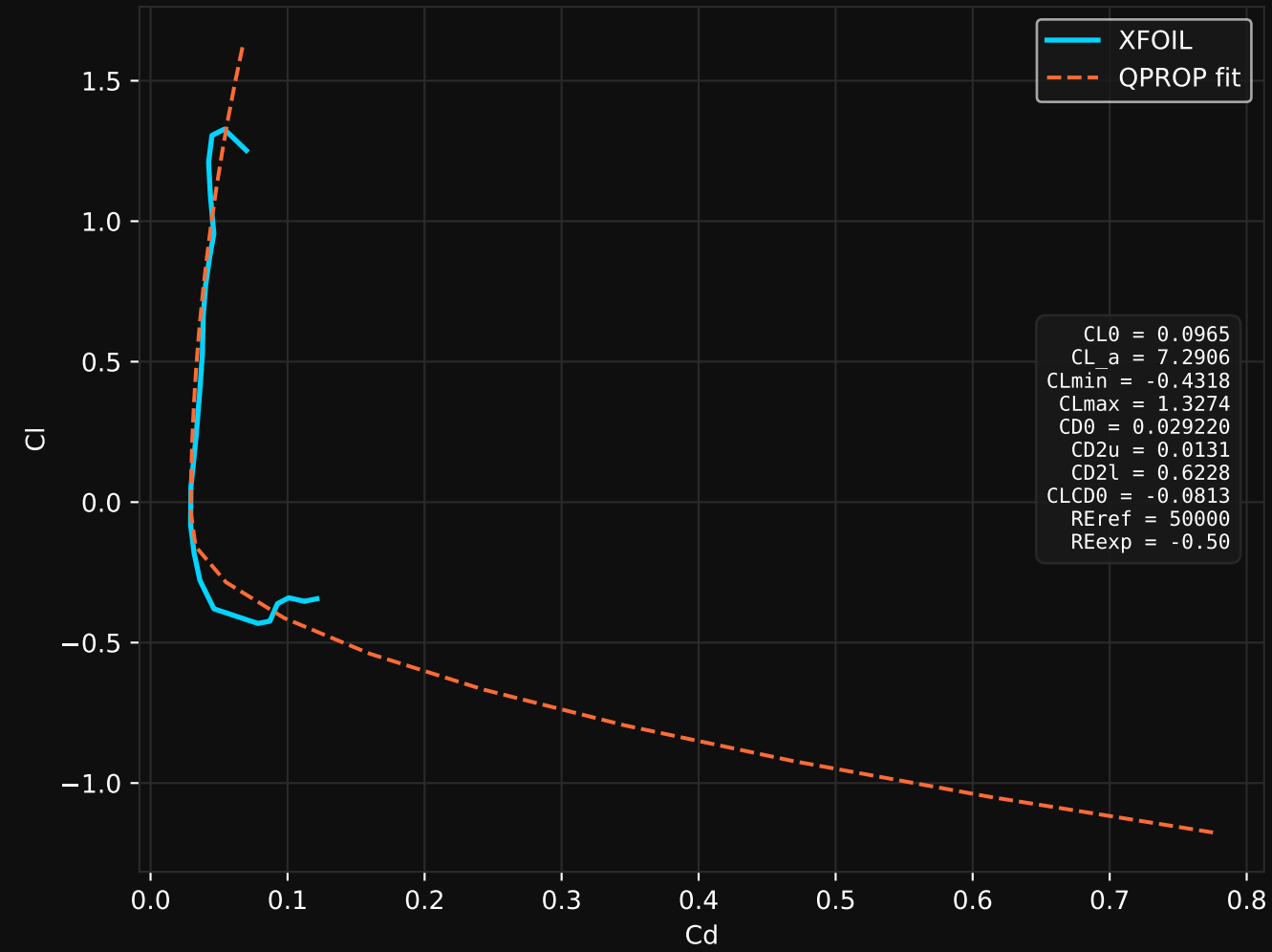


# Airfoil: ClarkY | Re = 50000

Cl x Alpha

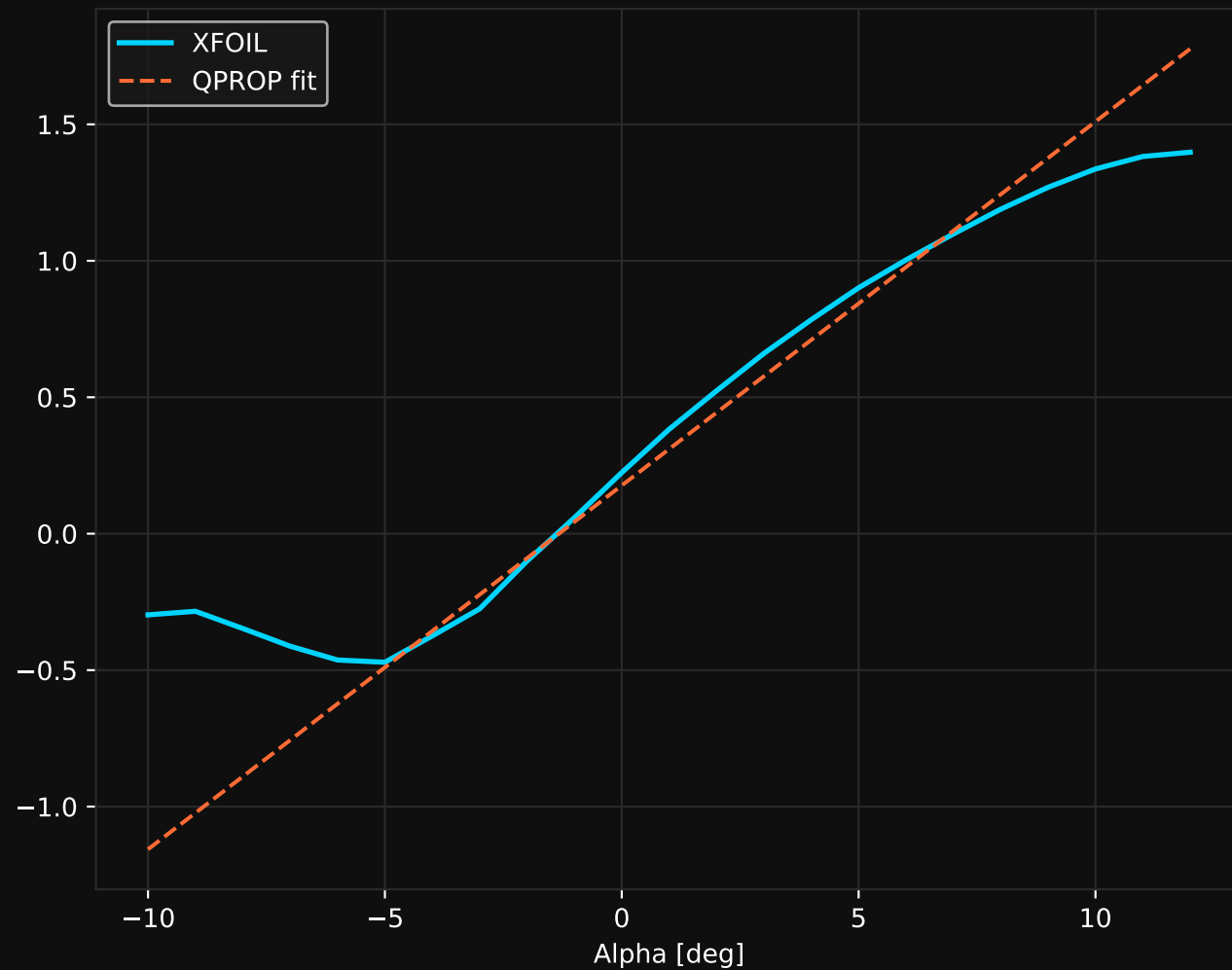


Cl x Cd (polar)

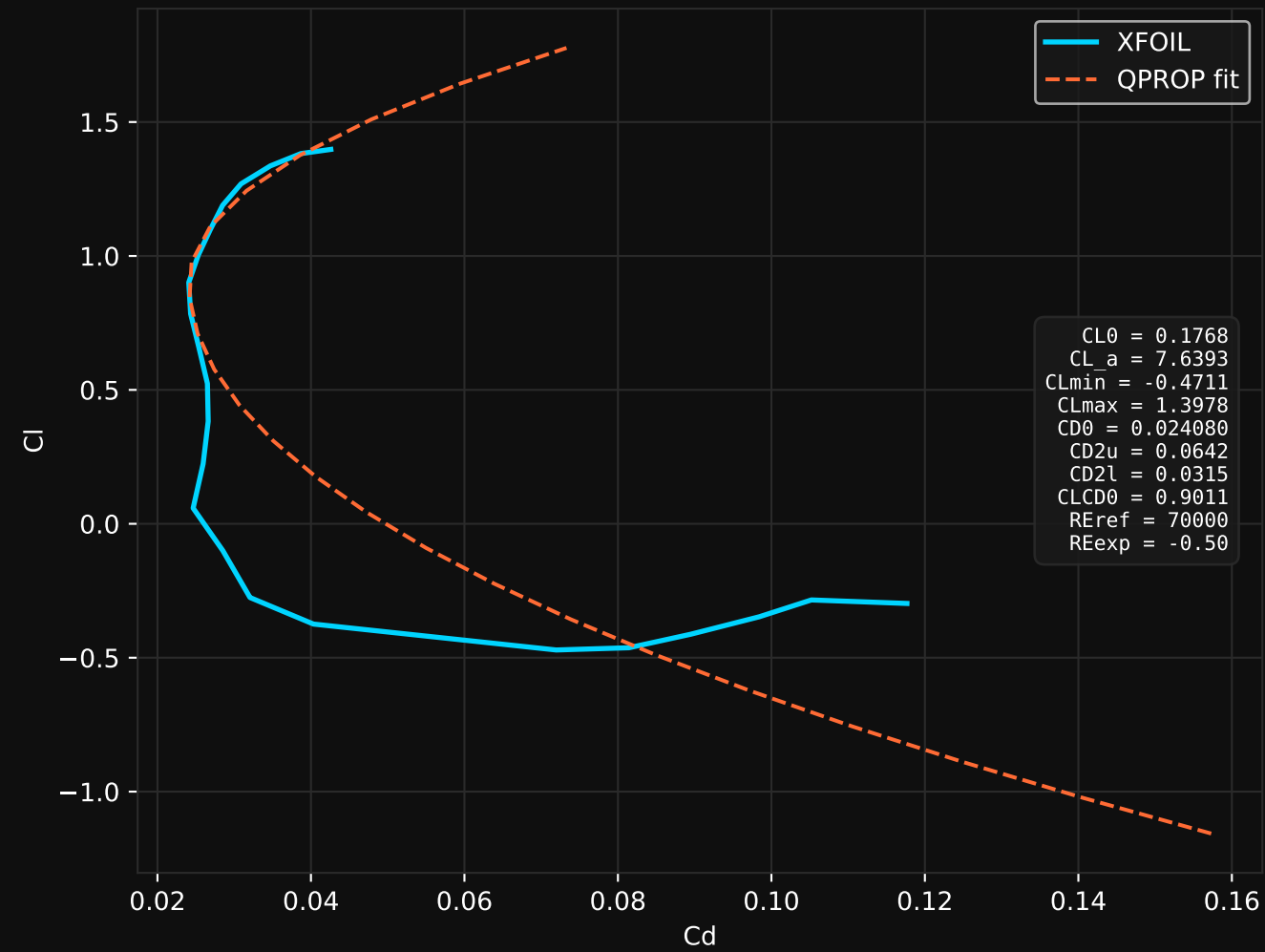


# Airfoil: ClarkY | Re = 70000

Cl x Alpha



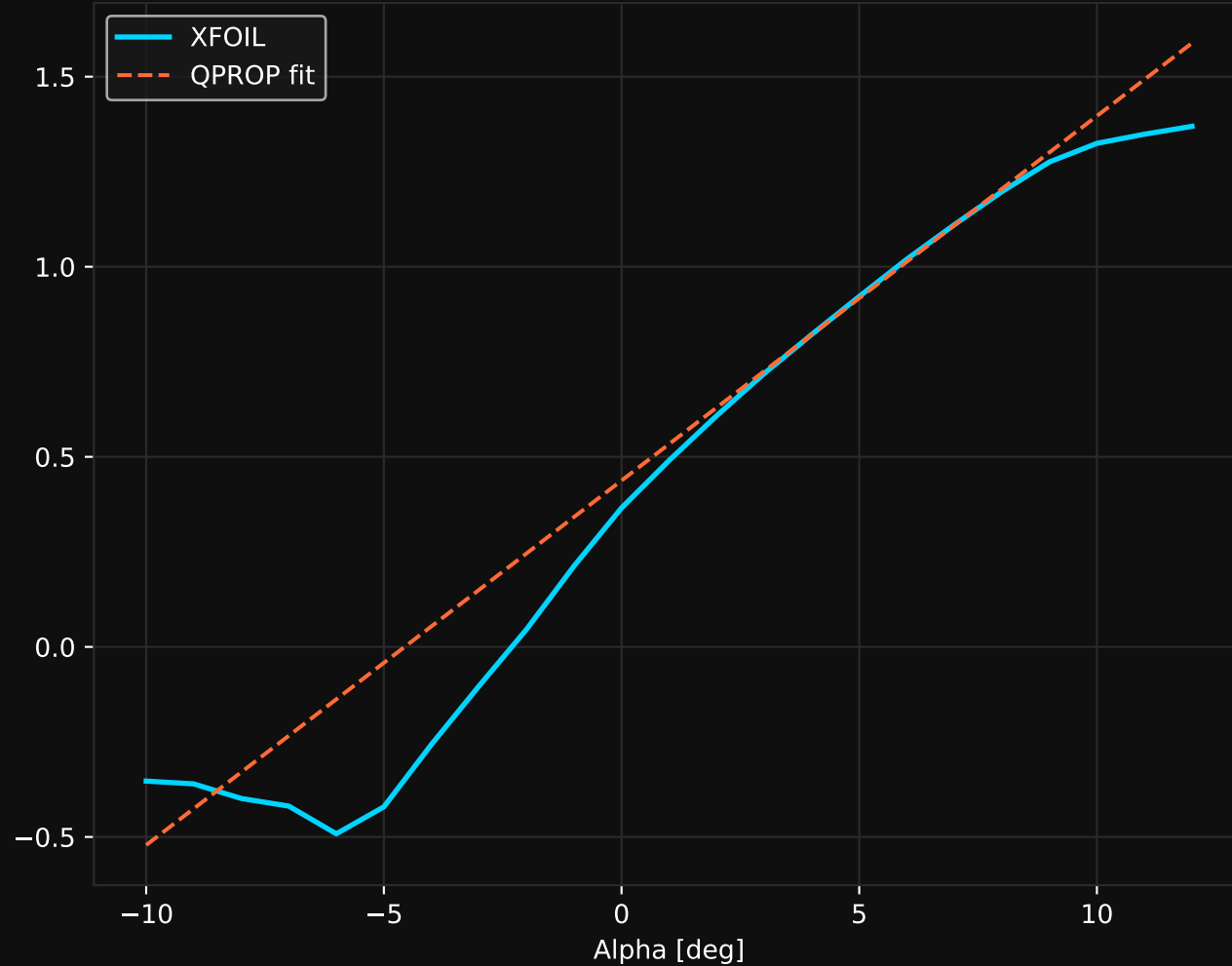
Cl x Cd (polar)



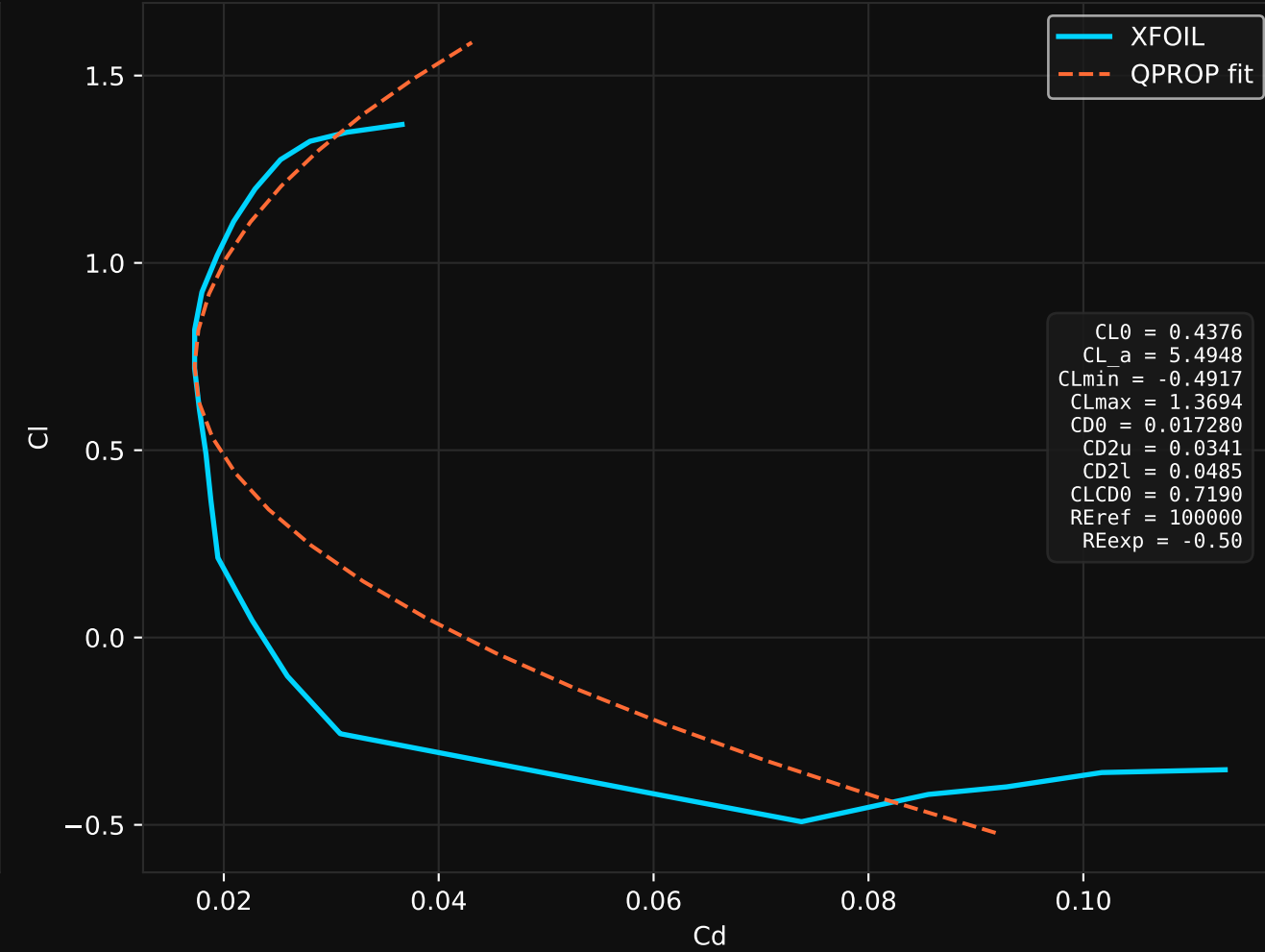
$CL_0 = 0.1768$   
 $CL_a = 7.6393$   
 $CL_{min} = -0.4711$   
 $CL_{max} = 1.3978$   
 $CD_0 = 0.024080$   
 $CD_{2u} = 0.0642$   
 $CD_{2l} = 0.0315$   
 $CLCD_0 = 0.9011$   
 $Re_{ref} = 70000$   
 $Re_{exp} = -0.50$

# Airfoil: ClarkY | Re = 100000

Cl x Alpha



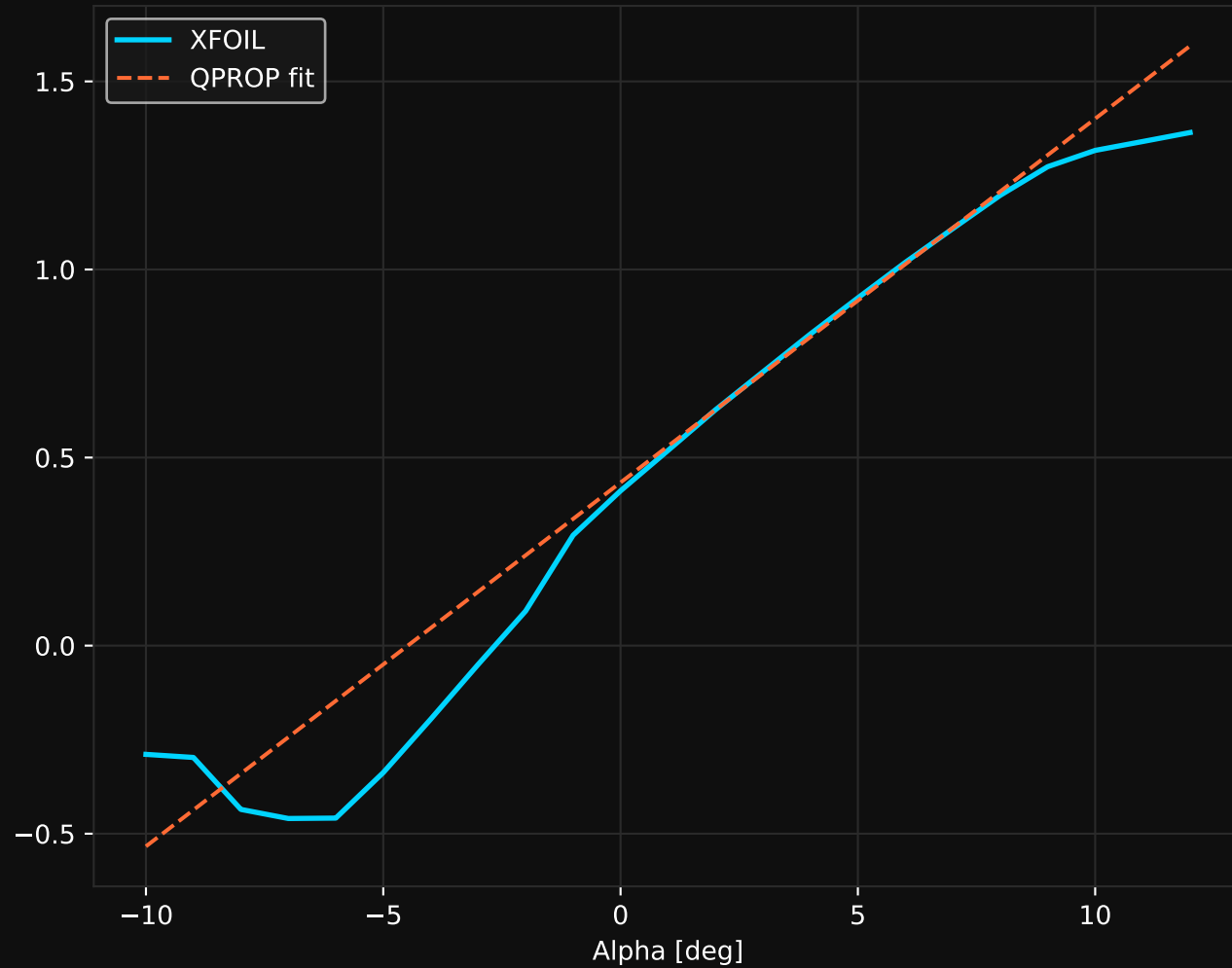
Cl x Cd (polar)



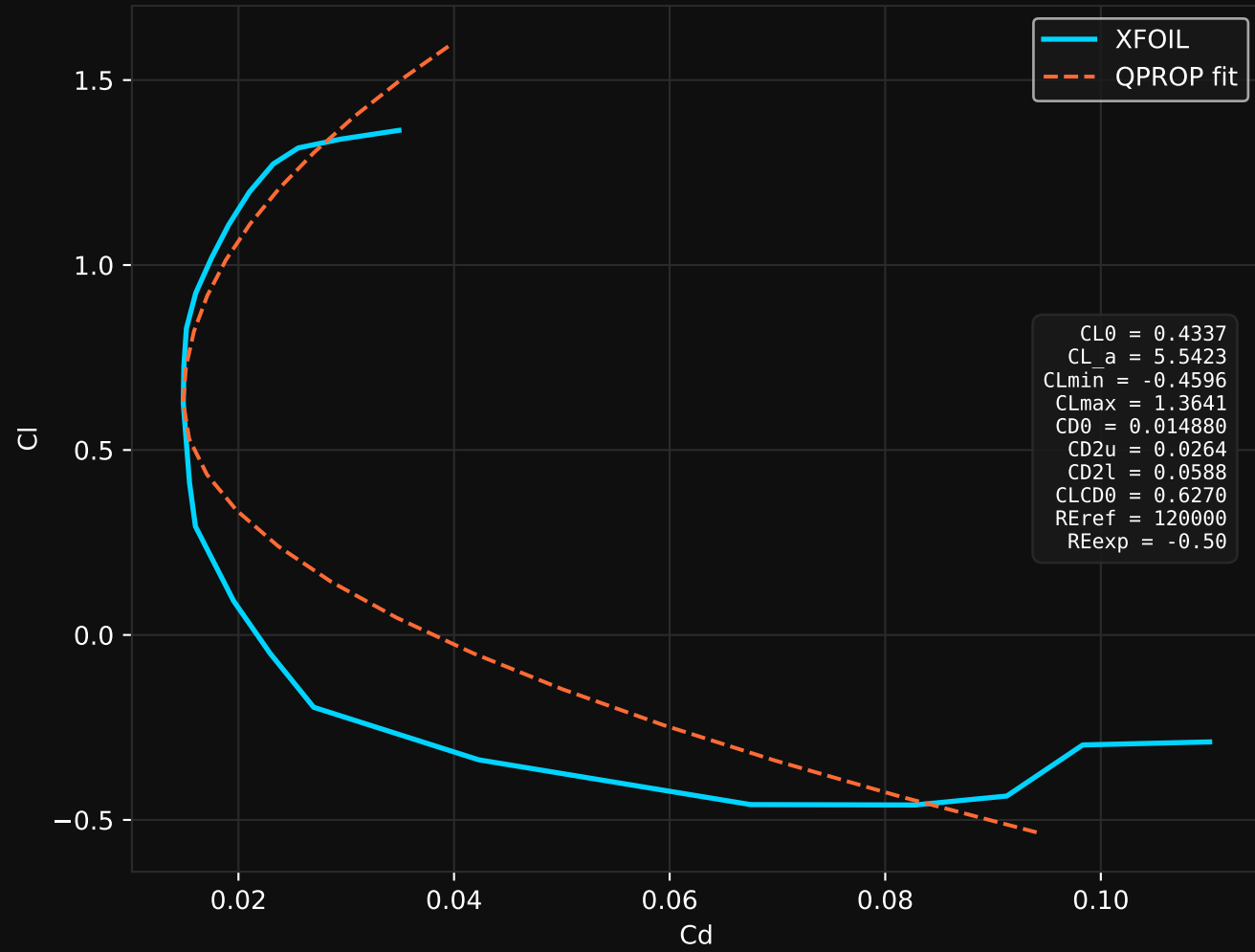
CL0 = 0.4376  
CL\_a = 5.4948  
CLmin = -0.4917  
CLmax = 1.3694  
CD0 = 0.017280  
CD2u = 0.0341  
CD2l = 0.0485  
CLCD0 = 0.7190  
REref = 100000  
REexp = -0.50

# Airfoil: ClarkY | Re = 120000

Cl x Alpha



Cl x Cd (polar)

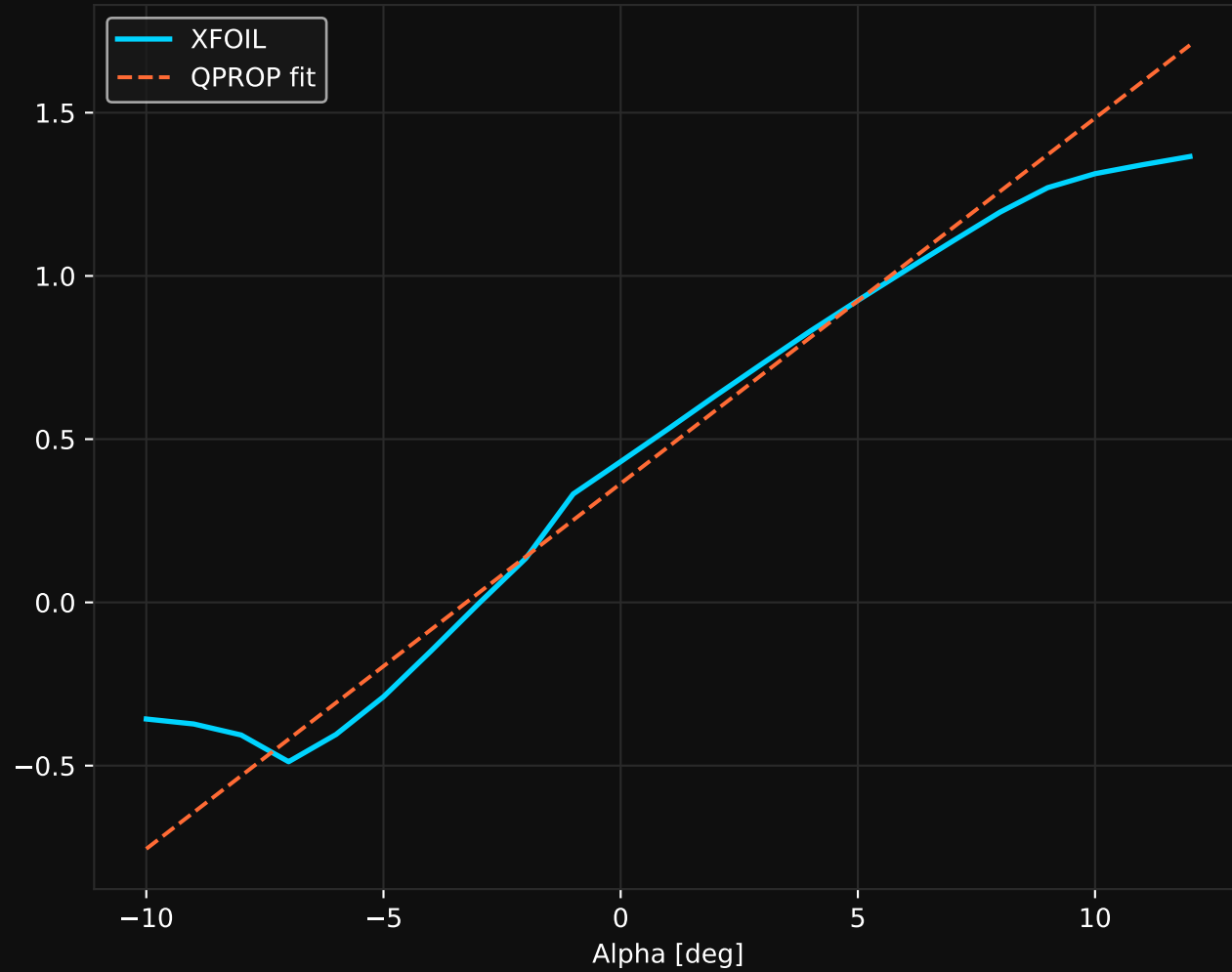


CL0 = 0.4337  
CL\_a = 5.5423  
CLmin = -0.4596  
CLmax = 1.3641  
CD0 = 0.014880  
CD2u = 0.0264  
CD2l = 0.0588  
CLCD0 = 0.6270  
REref = 120000  
REexp = -0.50

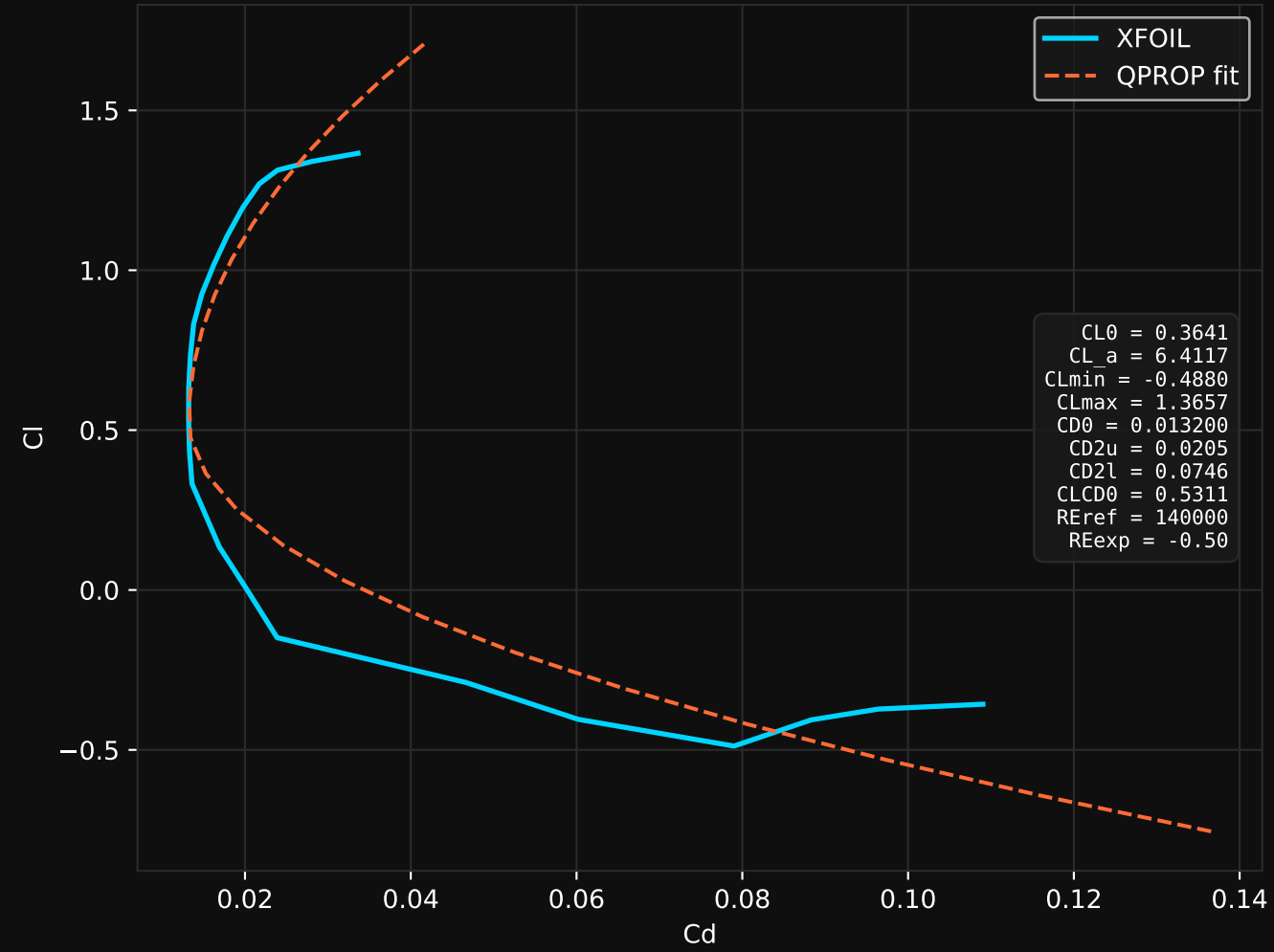


# Airfoil: ClarkY | Re = 140000

Cl x Alpha

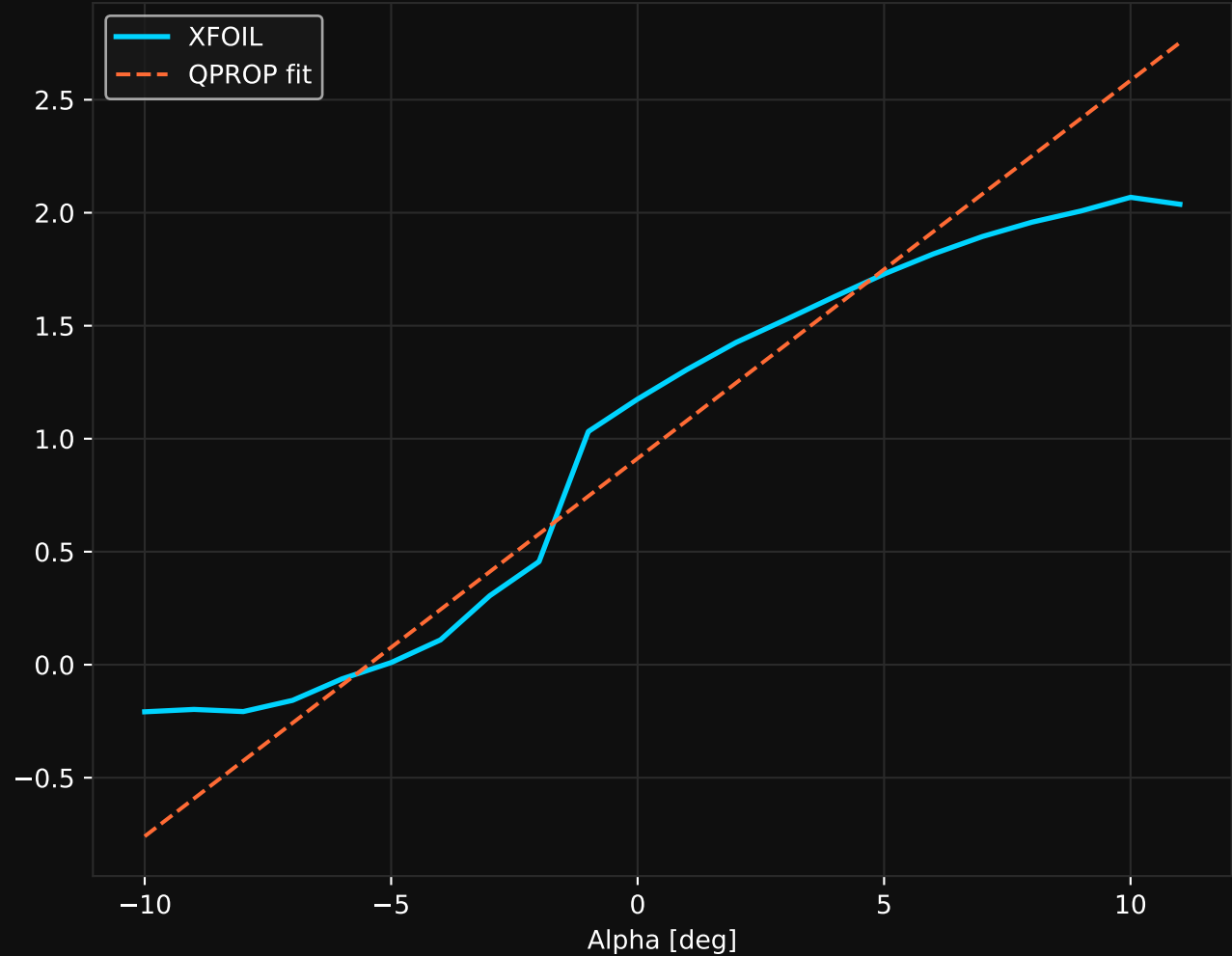


Cl x Cd (polar)

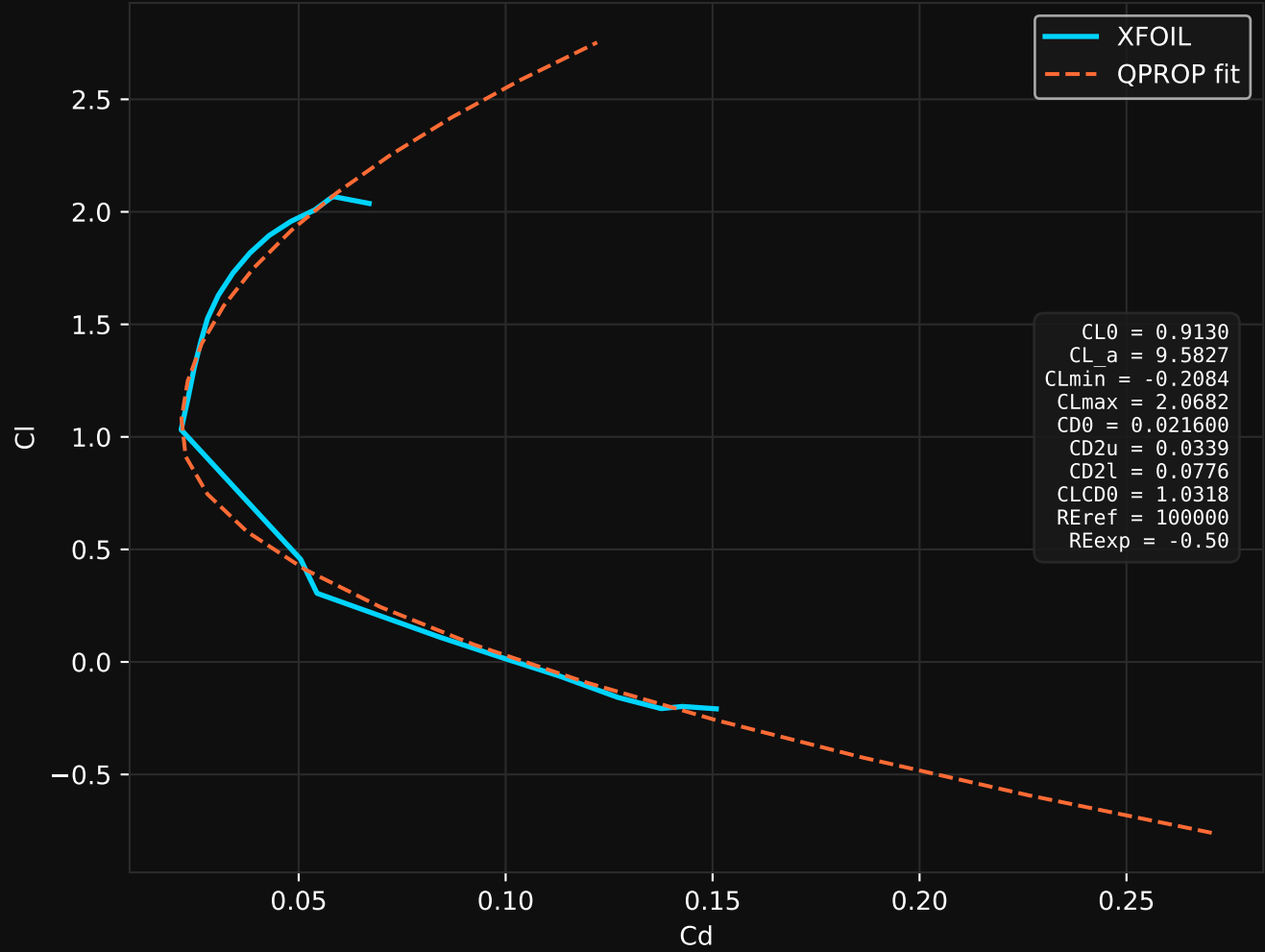


# Airfoil: s1223 | Re = 100000

Cl x Alpha



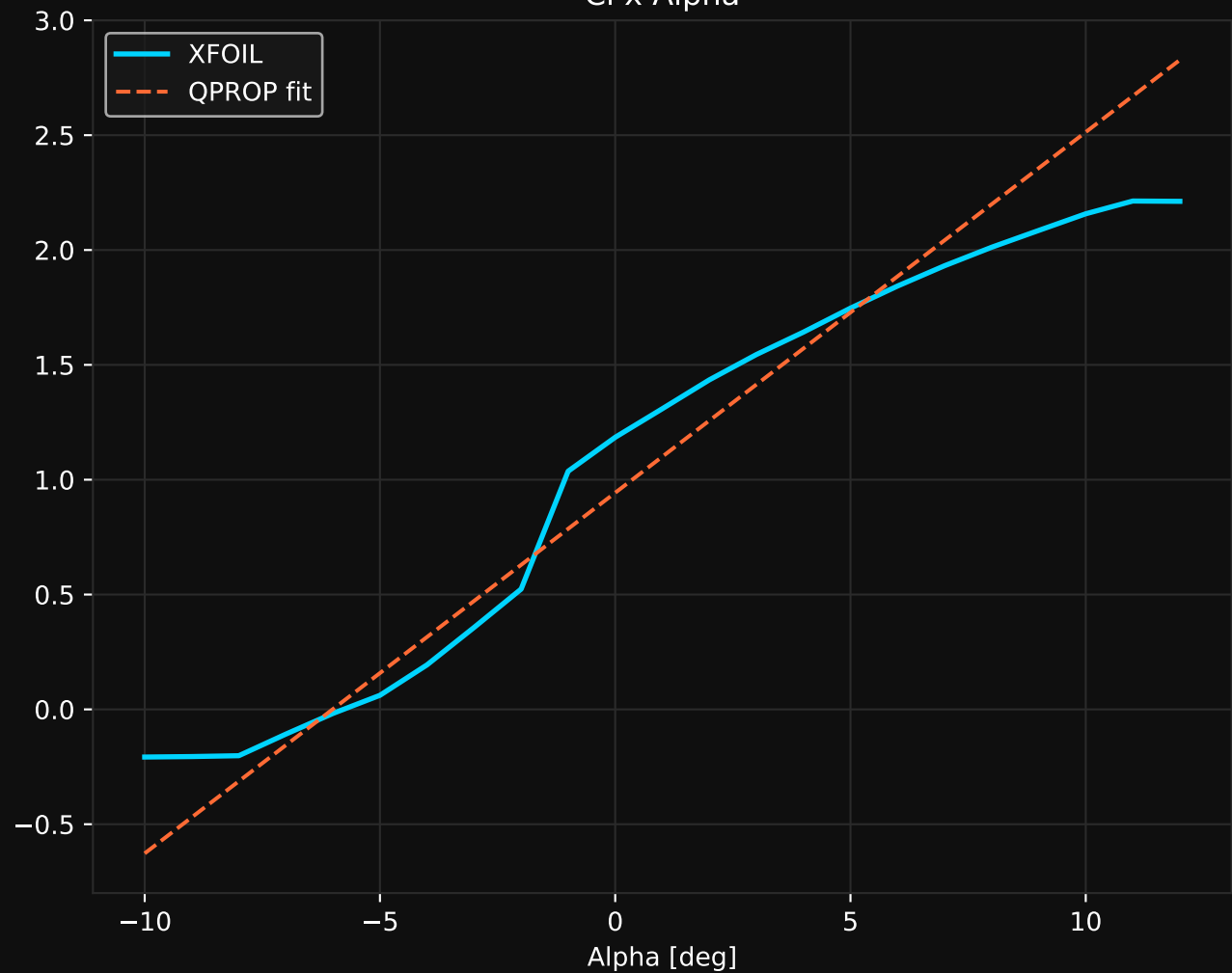
Cl x Cd (polar)



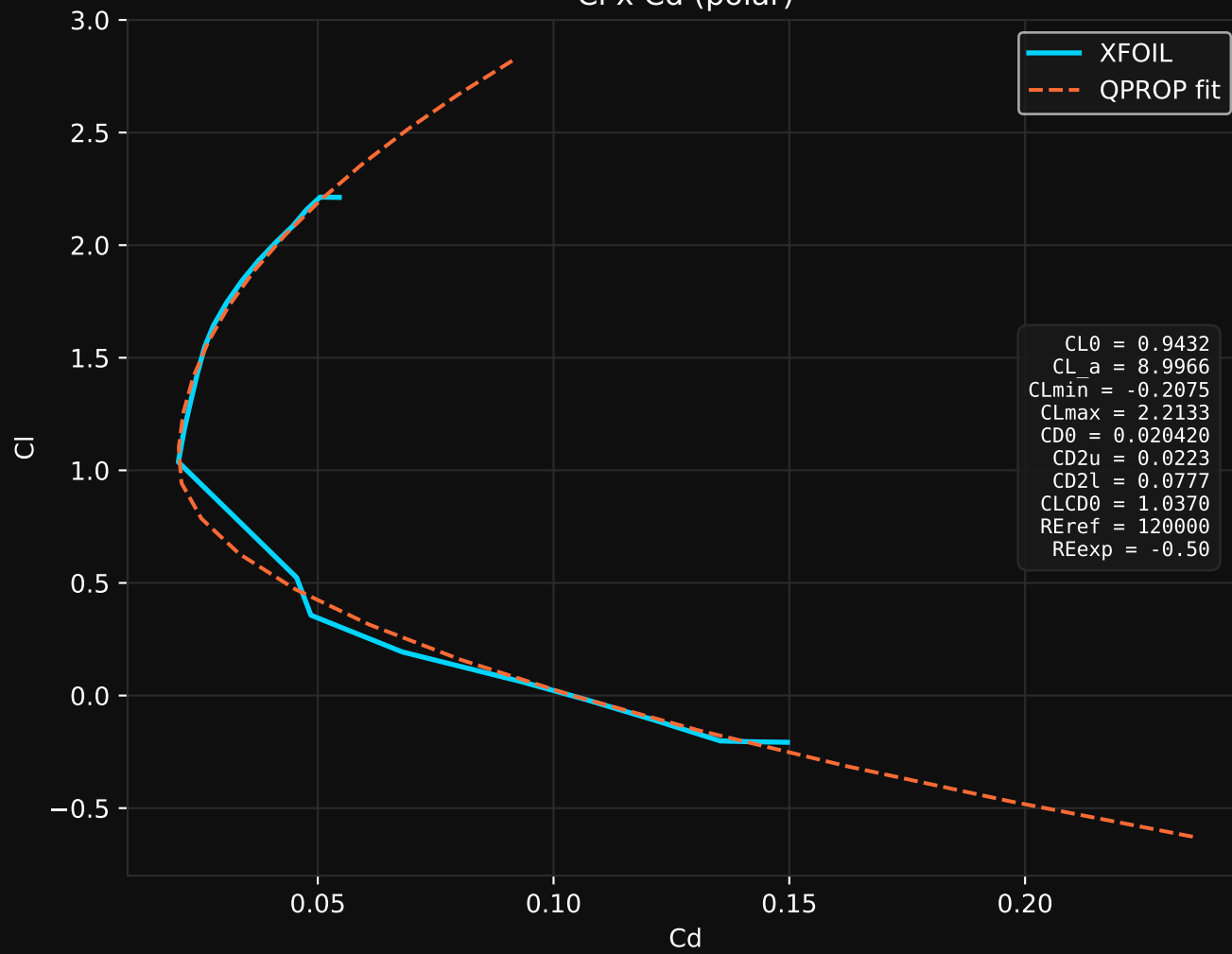
CL0 = 0.9130  
CL\_a = 9.5827  
CLmin = -0.2084  
CLmax = 2.0682  
CD0 = 0.021600  
CD2u = 0.0339  
CD2l = 0.0776  
CLCD0 = 1.0318  
REref = 100000  
REexp = -0.50

# Airfoil: s1223 | Re = 120000

## Cl x Alpha



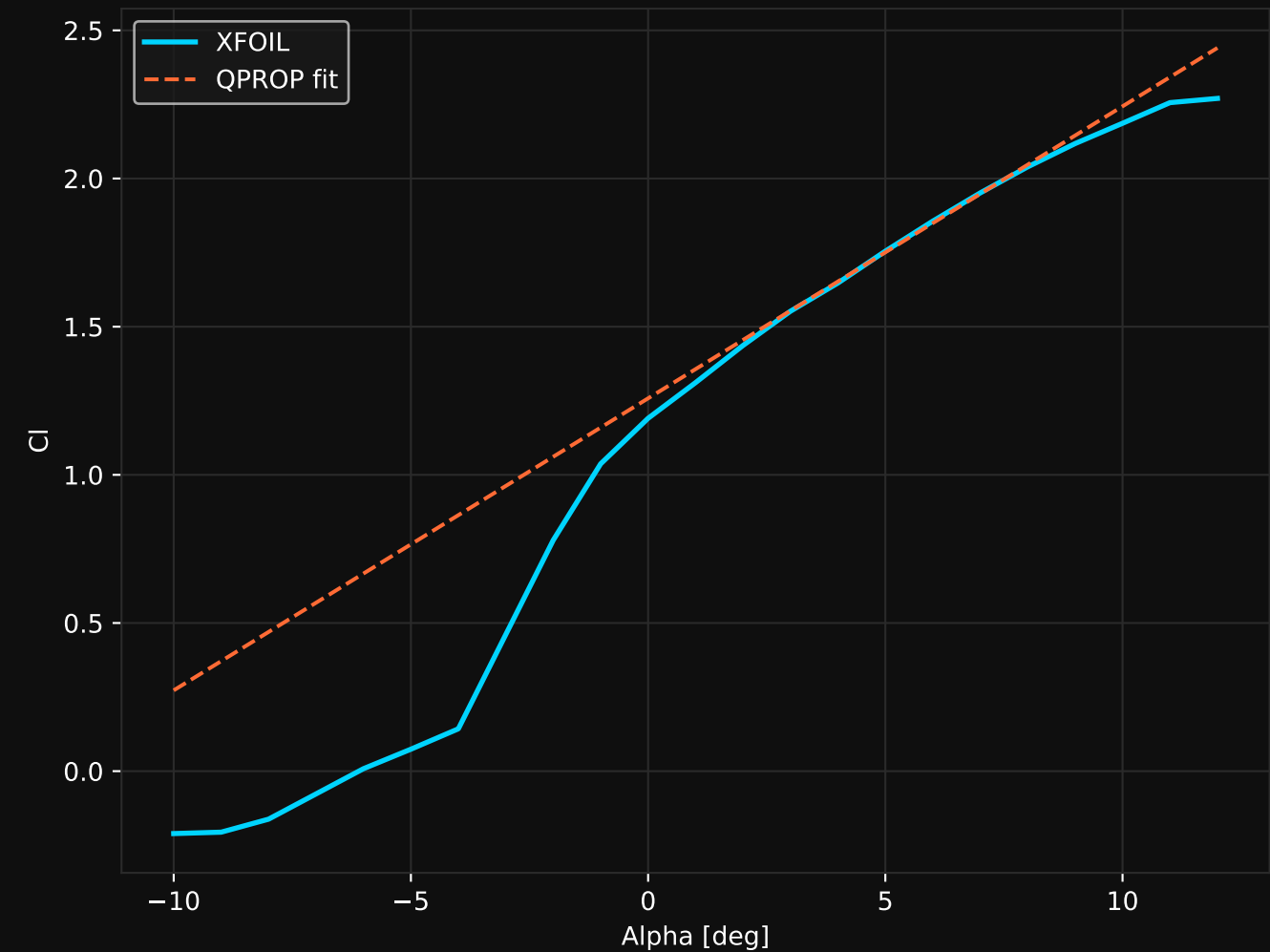
## Cl x Cd (polar)



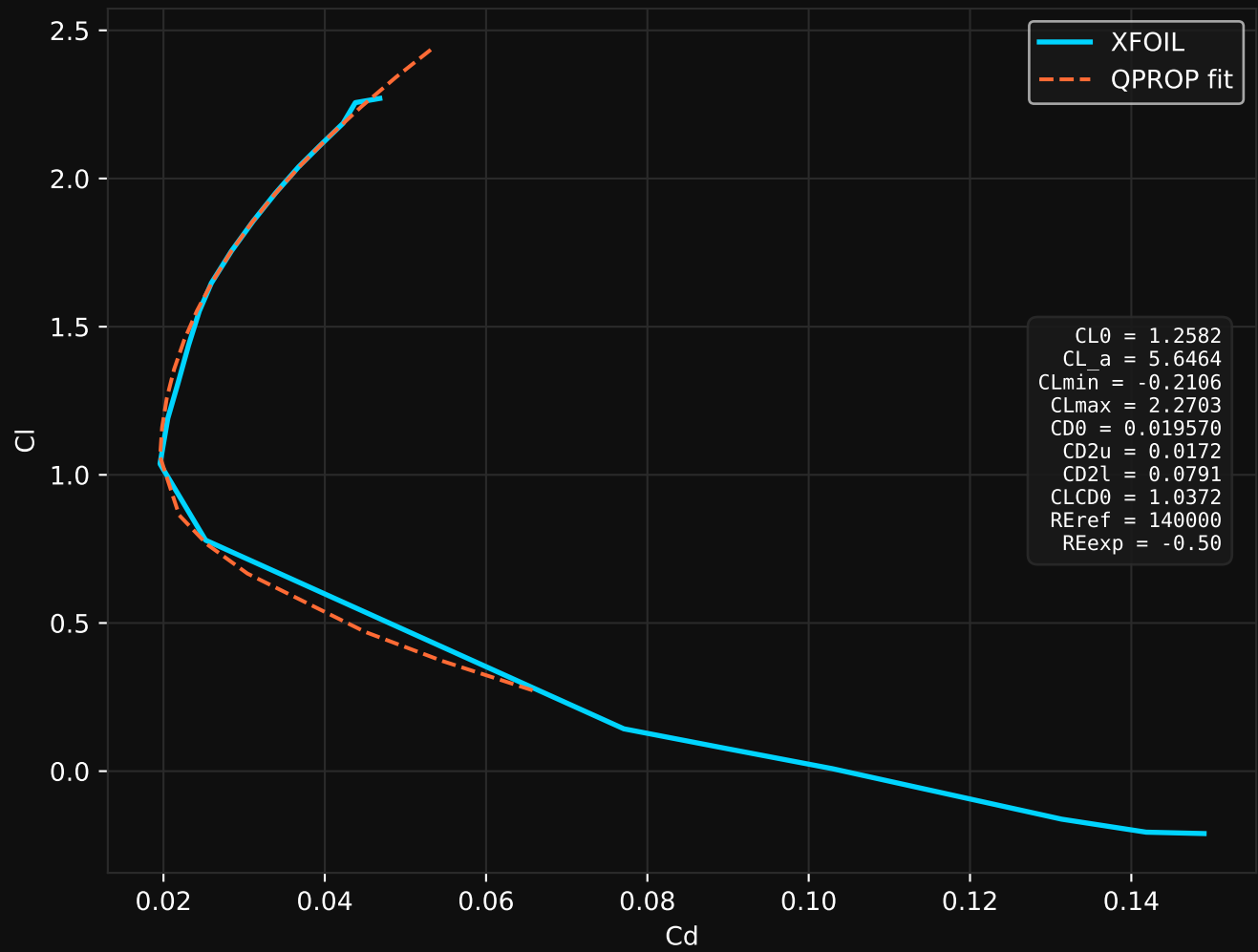
$C_{l0} = 0.9432$   
 $C_{l_a} = 8.9966$   
 $C_{lmin} = -0.2075$   
 $C_{lmax} = 2.2133$   
 $C_{D0} = 0.020420$   
 $C_{D2u} = 0.0223$   
 $C_{D2l} = 0.0777$   
 $C_{LCD0} = 1.0370$   
 $RE_{ref} = 120000$   
 $RE_{exp} = -0.50$

Airfoil: s1223 | Re = 140000

Cl x Alpha



Cl x Cd (polar)



$C_{l0} = 1.2582$   
 $C_{l\alpha} = 5.6464$   
 $C_{lmin} = -0.2106$   
 $C_{lmax} = 2.2703$   
 $C_{D0} = 0.019570$   
 $C_{D2u} = 0.0172$   
 $C_{D2l} = 0.0791$   
 $C_{LCD0} = 1.0372$   
 $Re_{ref} = 140000$   
 $Re_{exp} = -0.50$